FINAL REPORT

Comparative Life Cycle Assessment of Rescued Bread Flour

Prepared by 6M Sustainability Consultants for Rescued NZ

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EXECUTIVE SUMMARY

This life cycle assessment investigates environmental impacts of Rescued NZ's rescued bread flour in a comparison with conventional wheat flour. The two products are compared on a basis of supplying 1 tonne of baking flour for baking purposes, in the Auckland region (New Zealand). Rescued NZ provided primary data for the rescued bread flour production process, and secondary data were retrieved from scientific literature and the ecoinvent 3.8 database.

The study finds that the rescued bread flour has lower environmental impacts in all five selected impact categories: carbon footprint, land use, fossil fuel use, freshwater ecotoxicity, and air pollution (PM 2.5). The effect of uncertainty in the underlying data was assessed using Monte Carlo simulation and it showed that there was high certainty in this result. Sensitivity analysis also showed that the results were robust to changes in some of the parameters associated with major assumptions for the rescued bread flour production process. The study also highlights the environmental hotspots from the production line: the flash spiral dryer (for carbon footprint and fossil fuel use) and the paper bag used to store the final product (for land use). To enhance the environmental performance of the rescued bread flour, the study gives its recommendations in its conclusion.

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CO_2	Carbon Dioxide
FSD	Flash Spiral Dryer
GHG	Greenhouse Gas
ILCD	International Reference Life Cycle Data System
kWh	kilowatt-hours
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
NZ	New Zealand

Introduction

Food waste is a global issue, costing billions of dollars in lost profits and management costs, largely due to unsold food from supermarkets and other vendors. Although some organisations recover and redistribute unsold food, a significant portion, approximately 23% in NZ, still ends up in landfills (Mirosa, 2023). As food decomposes, it emits methane, a major contributor to greenhouse gas (GHG) emissions, while the resources used in its production, including land, water, fertilizers, and energy, are wasted. Beyond waste disposal, the agricultural sector contributes to environmental impacts through changes in land use, chemical inputs, water use, and emissions across the production, processing, transportation, storage, and retail stages. Addressing food waste, therefore, has both environmental and economic significance.

In NZ, a typical household wastes an estimated $1,510 worth of food annually, with vegetables (38%) and bread (29%) being the most wasted foods (Kantar, 2024), Rescued NZ is a NZ start-up established in October 2021, a start-up founded in October 2021 by Diane Stanbara and Royce Bold, upcycles unsold bread and other products. According to the Upcycled Food Association (2020, p. 8), upcycled foods utilize ingredients that would otherwise be discarded, are produced through verified supply chains, and have a positive environmental impact. In 2024, Rescued NZ diverted over 100 tonnes of waste, sourcing around 60% of its ingredients from food banks and local growers (Sima et al., 2025). By upcycling, the company helps reduce agricultural waste and landfill disposal, thereby lowering environmental impacts while creating economic benefits.

This LCA presents a comparative study of one of their innovative upcycled products, the rescued bread flour, and a conventional wheat flour imported from Australia (wheat flour). Diane and Royce have partnered with 6M Sustainability Consultants to carry out this study, aiming to provide credible, science-based evidence of their product's carbon footprint. This study assessed energy use, emissions, and other potential environmental impacts, offering insights that can help inform both stakeholders and customers about the true sustainability benefits of their rescued bread flour.

Overview of LCA

LCA provides objective, robust, and well-documented evidence of a product's potential environmental impacts. It takes into consideration all the processes required to deliver the product, from the extraction of raw ingredients through production, transport, use, and waste management. LCA results provide a quantitative and qualitative interpretation of the processes involved in the product's life cycle, according to the chosen environmental impact categories.

LCA serves multiple purposes, including the comparison of product systems, identification of environmental hotspots, and provision of evidence for policy and marketing decisions. It enables the relevant, transparent, and objective results when conducted under established standard. This assessment is carried out in accordance with ISO 14040:2006 and ISO 14044:2006, and it follows the guidelines of the International Reference Life Cycle Data System (ILCD) Handbook (European Commission, Joint Research Centre, 2010).

The ISO 14040 defines four phases for an LCA report: (1) goal and scope definition, (2) life cycle inventory (LCI), (3) life cycle impact assessment (LCIA), and (4) result interpretation. All of these phases follow a defined checklist to ensure that relevant information, assumptions, and limitations are transparently disclosed and documented. This report adheres to these standards to maintain transparency and consistency. The study's scope is set to enable a fair comparison between Rescued NZ's rescued bread flour and conventional wheat flour imported from Australia.

1. Goal Definition

The goal definition is the first phase of this comparative LCA study, defining and describing the purpose and intended use of the study in detail.

1.1 Intended Application of the Results

The results of this study are intended to be applied in three ways:

Identifying environmental hotspots

The study will quantify the environmental performance of rescued bread flour, with a particular focus on the carbon emissions. A selection of the relevant impact categories for the study will reveal the processes that are the primary contributors to potential environmental degradation, the environmental hotspots.

Comparative assessment

The study will compare the environmental impacts of rescued bread flour and wheat flour. This comparison will provide qualitative, quantitative and objective insights between the processes of upcycling surplus bread and importing ingredients.

Marketing purposes for Rescued NZ stakeholders

As an innovative and growing company, Rescued NZ aims to inform its existing and potential stakeholders, collaborators, and customers about the scientific evidence supporting its rescued bread flour, thereby enhancing their efforts in upcycling food. Hence, the LCA results can be used internally for marketing purposes to communicate to various stakeholders.

1.2 Limitations Due to Methodological Choices

The limitations of this LCA study arise from several methodological choices and constraints:

Software limitations

The study uses openLCA software, incorporating the ReCiPe2016 LCIA method and the ecoinvent 3.8 database to assess environmental impacts. As some data are not represented (e.g., average processes in NZ are often not represented), alternative datasets from other countries are used based on justified assumptions. All such assumptions are clearly documented to ensure transparency in the LCI data selection.

Selection of impact categories

Each LCA software defines its own impact categories, and this study interprets only those relevant to its intended application. Some categories are excluded to focus on areas with potential environmental hotspots. Results are presented as impact scores, which may be challenging for general readers to interpret. Therefore, the report includes explanations and definitions to enhance understanding, although some terms may still require further clarification.

Wheat flour and rescued bread flour environmental impact differences

Selecting appropriate impact categories can be challenging when comparing two products. For instance, wheat grain production is typically associated with impacts such as energy consumption, human toxicity, terrestrial ecotoxicity, acidification, and eutrophication (Narayanaswamy et al., 2004), whereas upcycled food studies often focus on land use and water footprint (Thorsen et al., 2024). In both cases, global warming and carbon emissions are equally important. The study will therefore focus on the most relevant impact categories to ensure a fair comparison between the two products.

Specificity of wheat flour and rescued bread flour differences

The two products differ in their production specificity; for instance, both products are produced at different locations (e.g., Australia vs. NZ), using different technological processes (e.g., crop farming vs. upcycling bread), and timeframe aspects (e.g., wheat farming is a longer process than upcycling bread). To ensure a fair comparison, assumptions and methodology are adjusted, focusing on the most relevant impact categories while acknowledging that some impact pathways may be simplified or omitted.

1.3 Decision Context and Reasons for Conducting the Study

This study is being carried out to generate an estimate of the environmental impacts of rescued bread flour on its own and relative to wheat flour to provide Rescued NZ and its stakeholders with reliable data to assess the impacts of upcycling food. Rescued NZ and its stakeholders intend to use the results of this study to potentially make decisions about product development, process design, or funding.

According to ISO 14040, ISO 14044 and ILCD guidelines, the decision context of this study is classified as Situation A (Micro-level decision support), as the results will support decision-making, but not result in structural changes.

1.4 Target Audience

The primary target audience for this study is Rescued NZ. The company may provide this LCA report to its supply chain and industry partners (e.g., KiwiHarvest), stakeholders (e.g., the Ministry for the Environment), and customers. Rescued NZ is responsible for taking ownership of the results and their communication. Hence, this LCA is written for an audience with limited prior experience in LCA.

The study is being conducted by 6M Sustainability Consultants from the University of Auckland as part of a group project for the course ENVENG752 (Sustainability and Life Cycle Assessment). Dr Febelyn Reguyal, Chief Executive Officer of 6M Sustainability Consultants, serves as the expert technical advisor for the project.

1.5 Comparisons to be Disclosed to the Public

This study is not intended to be a comparative assertion disclosed to the public and is not subject to external review or legal requirements.

1.6 Commissioner of the Study and Other Influential Actors

The study is fully commissioned by Diane Stanbra and Royce Bold, co-founders and co-directors of Rescued NZ. They have defined the study's goal and scope in collaboration with 6M Sustainability Consultants (6M). The company is responsible for providing operational data and data from its supply chain partners when possible. They are taking full ownership of the results and their communication to the stakeholders.

2. Scope Definition

2.1 Deliverables

The deliverables of this study are:

- Executive Summary for a non-technical audience
- The LCA main report including:
    - Goal and scope
    - LCI
    - LCIA
    - Interpretation
- A list of appendices documenting raw data, assumptions, and full LCI results
- A visual presentation (video and poster) to the commissioner

There will be no separate report for internal use, as no confidential information will be used in this LCA.

2.2 Object of the Assessment

The objective of the LCA study is to compare wheat flour with rescued bread flour used in the Auckland Region. The following information was used to inform the function and functional unit of the study:

- Information regarding the logistics and partnerships:
    - Rescued NZ received the surplus bread from KiwiHarvest, their partner, which collects it from products from Auckland region supermarkets (e.g., Woolworths, New World, Costco). According to Rescued NZ documentation, Woolworths is the largest supplier.
    - Rescued NZ partners up with The Foodbowl, a commercial food manufacturing facility in Māngere, where they produce the rescued bread flour.
    - Australia is the main importer of wheat grains in NZ. It is mainly used for the dairy sector to feed the pasture. Queensland, Victoria, and New South Wales regions are the main producers and exporters of wheat (White et al., 2018). The study considers the geographical location "Australia" as the average distance of the three regions' ports to Tauranga port in NZ.
- Information regarding the process and product specificity:
    - The surplus bread is produced in a large-scale commercial kitchen from supermarkets. There is a variety of bread produced with different choices of ingredients, sources of ingredients and processes used. Hence, this study considers the surplus bread as a typical bread loaf found in NZ supermarkets. However, the object of assessment should be focused on the rescued white bread flour product, only from surplus white bread loaves (as opposed to grain or wheat that they also produce).

Figure 1 represent the journey of the production of both products.

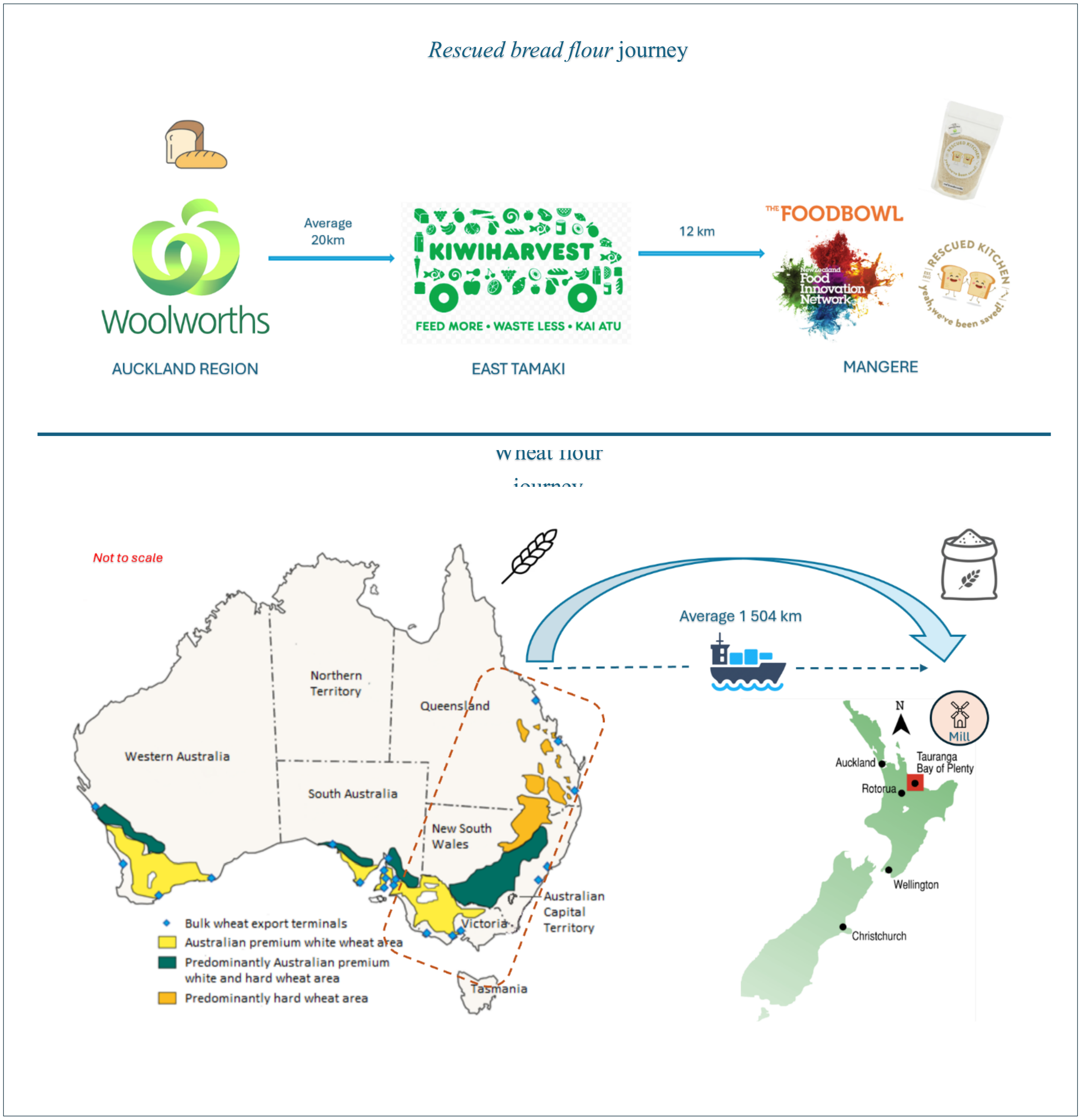

Figure 1 - Rescued bread flour and wheat flour journey (source: Hertzler et al. (2013) and Worldmap)

2.3 Function, Functional Unit, and Reference Flow

The function, functional unit and reference flow define the parameters for the intended application. The information is essential for framing the object of the assessment, both qualitatively and quantitatively, and serves as a foundation for the collection of data for the LCI modelling.

2.3.1 Function

Rescued bread flour is designed to fulfill the same core function as wheat flour, providing comparable baking properties that make it suitable for preparing a wide variety of flour-based products, such as loaves, cakes, and muffins. Its strict production ensures allergen control by sourcing from the bread loaves that exclude sesame seeds and nuts. Additionally, rescued bread flour is differentiated by its availability in multiple bread flour varieties (white, wheat, or wholegrain), and it maintains similar tastes to conventional wheat flour

2.3.2 Functional Unit

The functional unit defines the quantitative and qualitative aspects of the function. It needs to reflect both products equally, hence this study established the functional unit as being:

Supplying 1 tonne of baking flour for baking purposes, in the Auckland region (NZ), stored in ambient room temperature.

To ensure that both products are compared within the same scope:

- Both flours are distributed and used in the Auckland Region.
- Both flours are used interchangeably for baking purposes, while they might differ in their ratio.
- Both flours are stored in Auckland in a similar storage condition to preserve the quality and shelf-life of the product.

2.3.3 Reference Flow

The reference flow (RF) is the amount of a product used to fulfill the study's functional unit:

- For rescued bread flour, it is based on the mass of bread loaves required to supply 1 tonne of flour, 1644 kg. This quantity reflects the information provided by Rescued NZ for processing the bread.
- The reference flow for wheat flour is based on the quantity of wheat grains to supply 1 tonne of flour, 1282 kg. This quantity assumes that there is some waste and the total yield outcome of wheat flour is 78% (New Zealand Flour Millers Association, n.d).

2.4 Secondary Functions and Handling of Multifunctional Processes

In LCA practice, if a product has multiple functions, the practitioner must determine how to treat the secondary functions. In this study, secondary products in both systems are handled using mass allocation.

For example, in this study, a byproduct of rescued bread flour is breadcrumbs which serves as a secondary product. Thus, the secondary function (breadcrumbs) will be treated via mass allocation, by simply removing the breadcrumbs mass from the overall results. The breadcrumbs have the exact same components as rescued bread flour; their particles are bigger in size.

2.5 LCI Modelling Framework: Attributional

This study examines the potential environmental impact of rescued bread flour and wheat flour, attributing the impacts to the processes and resources required to produce these two flours. This study is an attributional LCA.

2.6 System Boundaries and Completeness Requirements

The system boundaries define the boundaries of the two products under review. Ideally, in LCA practice, it is preferred to examine the entire life cycle of a product (cradle-to-grave), from raw material extraction (e.g., grain farming) to end-of-life (e.g., flour might end up in a landfill if not consumed). In this study, the commissioner requested a cradle-to-gate approach. The term "cradle" refers to the raw material extraction, which identifies the source of the raw material to start the manufacturing process (e.g., the wheat grains in the wheat flour, and the surplus breads in the rescued bread flour). The "gate" refers to the gate of the storage facility before the flours are transported to retailers and customers (downstream processes, see definition in Table 1).

The two products have distinct system boundaries and are being compared based on the same functional unit defined earlier (see Section 2.3.2); thus, they can be compared fairly. To understand the terms used in the illustration of the system boundaries, some definitions are given in the following Table 1.

Table 1 - Processes definition terms

Term	Definition
Downstream processes	It includes processes that occur after the manufacturing and packing phases. The downstream processes include distribution (transport to retailers), the use phase (consumption), and waste treatment (either landfill or possible recycling).
Core process	It refers to the production of a product, also known as the manufacturing stage or processing phase.
Upstream process	It includes the processes prior to the core processes (production stage), which involve the extraction of raw materials and the delivery of these raw ingredients for the processing phase.

2.6.1 Wheat flour system boundaries

Australia exports mainly wheat grains, a report revealed that only 10% of the wheat imported in NZ as wheat flour. This study considers that the wheat grains are imported from Australia and milled in Tauranga, NZ (United State Department of Agriculture, 2025). This information is supported by Rescued NZ.

The steps for wheat flours derived from a typical wheat flour production (Australian Export Grains Innovation Centre, 2024), it is resume in Table 2 and illustrated in Figure 2.

Table 2 - Typical wheat flour production process

Wheat flour processes
Start of process - Cradle
Step 1 – Farming Farmers prepare the paddock, sow wheat, and care for the crop. This includes getting the soil ready, controlling weeds and pests, adding water or nutrients when needed; then harvesting the wheat grains with a combine and store the grain safely before it leaves the farm.
Step 2 – Sourcing & Handling Wheat grains are processed for storage by cleaning, removing stones, chaff and dust. The wheat grains are then stored before transport.
Step 3 – Transport to NZ Wheat grains are transported to an export hub or port by road and/or by train, then shipped by cargo boat to Tauranga port. The port has a mill factory on its premises.
Step 4 – Grading at mill intake The wheat grains are sampled and checked for basic properties and impurities, then graded. Batches that do not meet the target may be downgraded or rejected, which becomes waste. Clear grading keeps the flour quality consistent.
Step 5 – Milling in NZ Before milling, the wheat grains are cleaned and conditioned into kernels. A conveyor belt and a sifter gently separate the outer layers and produces flour. Bran and germ are collected and used for other purposes; these are the wheat flour byproducts.
Step 6 – Quality control The milled products are checked for quality control to meet its specification through x-ray or metal detection to remove any objects. The batches are weighed, ready to be packed
Step 7 – Packing and Delivery to Auckland The wheat flour leaves the mill either in bulk for silos, or in paper bags for bakeries and food factories They are labelled with their batch information for traceability. They are transported to Auckland following food safe transport practices.
Step 8 – Storage The final product is stored in an ambient room temperature in Auckland. For the study, it is assumed to be stored in the same conditions as the ***rescued bread flour.***
End of process - Gate

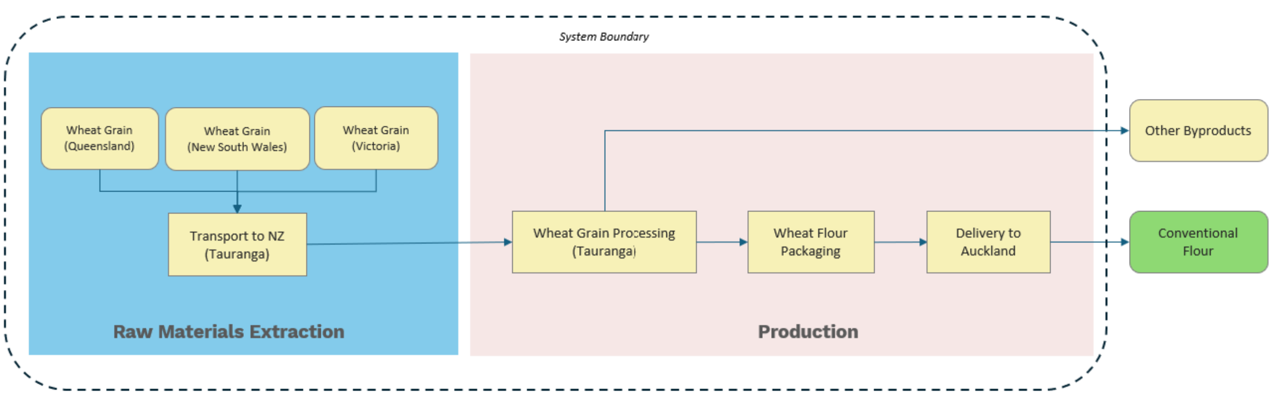

Figure 2 - Wheat flour system boundary

2.6.2 *Rescued bread flour* system boundaries

The steps described below are based on interviews with Rescued NZ co-founders and co-directors, Diane and Royce, and are summarised in the Table 3 and illustrated in Figure 3.

Table 3 - Rescued bread flour production process

Rescued bread flour processes
Start of process - cradle
Step 1 –Collection of surplus bread *KiwiHarvest* collects frozen surplus bread in refrigerated trucks from its suppliers, supermarkets in the Auckland region, and stores it in a freezer in its warehouse located in East Tamaki, Auckland.
Step 2 – Delivery of surplus bread *KiwiHarvest* delivers the surplus bread to the processing plant, located in Mangere, Auckland. Rescued NZ is partnering with *Foodbowl* to use their processing plant.
Step 3 – Debagging of surplus bread The surplus bread is still packed in its commercial soft plastic packaging and needs to be removed from its packaging. The soft plastic is recycled (out of scope for the system boundaries).
Step 4 – Drying of milled bread Rescued NZ is utilising a flash spiral drying (FSD) method to dry the bread. Once in the conveyor, the bread particles drop into the spiral dryer, which spins around by using the hot air that "flashes" into steam; it dries the milled bread loaf rapidly. As the bread particles dry, they become hotter and rise to the top, exit a funnel and become dried milled bread of various particle sizes. The outcome is approximately a 72% yield of dried milled bread, with a 28% loss due to the water content removed in drying. The output is bread pellets.
Step 5 – Sieving of bread pellets The bread pellets are then sifted using a vibrating sifter set at 0.5mm size. This separates the breadcrumbs (> 0.5mm), the *rescued bread flour* (< 0.5mm) The output of this process estimated to be 85% *rescued bread flour* and 15% breadcrumbs by mass.
Step 6 – Final Packaging The *rescued bread flour is* bagged using 15kg reusable paper bags.
Step 7 – Quality control: X-ray and weighing The *rescued bread flour* an X-ray check to detect any metal contaminants, which is a standard procedure for ensuring food quality. The products are then weighed and ready to be packed.
Step 8 – Storage The final products are stored in an ambient room temperature at the warehouse at *FoodBowl* until they are dispatched for future use.
End of process - Gate

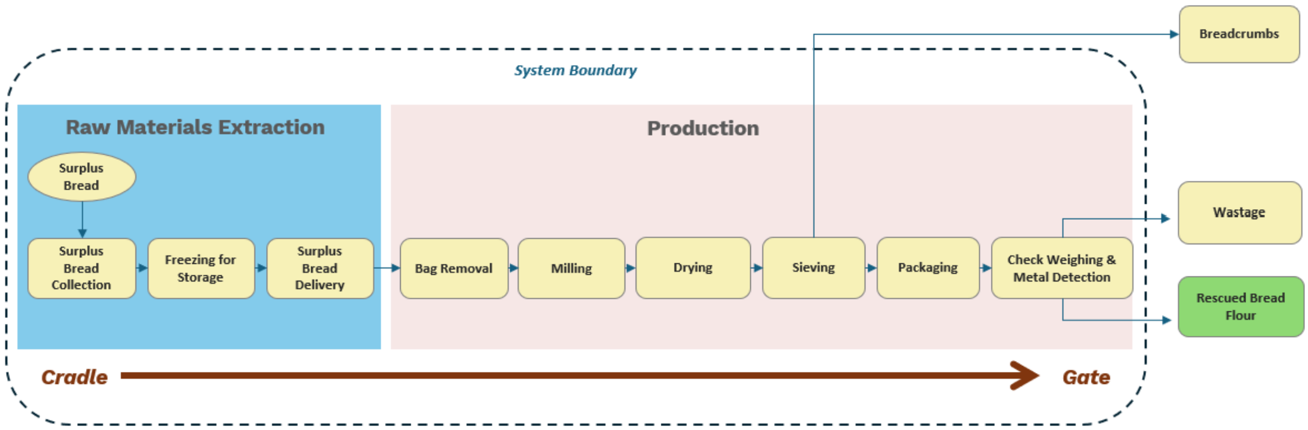

Figure 3 - Rescued bread flour system boundary

2.7 Components Excluded from the study

The study excludes a few components:

- For rescued bread flour: This assessment excludes the environmental impacts associated with the production of surplus bread loaves. Additionally, the surplus bread's soft plastic packaging is excluded from the baseline of this LCA.
- For wheat flour: Wheat farming involves several processes that may not occur at the same location (e.g., grain farming and bagging may not take place at the same farm, tractors are used to displace the grains); therefore, additional transportation may be required. The study excludes all transport used in the wheat farming process; only the transport from farm to port in Australia, shipping to NZ and within NZ are considered.
- For both products:
  - *Infrastructure*: It is common practice in LCA to exclude capital equipment involved in product production; hence, the dehydrating plant warehouse and the wheat farming warehouse are excluded from the analysis, along with all resources required for their operation.
  - *Support Functions*: Any organisation's supporting activities resources are excluded from the research (e.g., research and development, marketing, sales, finance, and business trips).
  - *Cut-off Criteria*: It is common practice in LCA to exclude materials (or ingredients) with a mass of less than 1% of the final product (e.g., dust and other volatile particles).

2.8 Data Assumptions and Limitations

There are several data assumptions and limitations associated with this study:

Source of data

Primary data collection for the wheat flour system is not feasible due to time and resource constraints. This study relies solely on the datasets available on ecoinvent 3.8 for the conventional wheat flour data. The data necessitate several assumptions that may impact the quality and confidence of the results. Therefore, secondary data from published studies and the ecoinvent 3.8 database are used to represent the processes. This difference in data quality and coverage may lead to uncertainty and limit the direct comparability of environmental impacts between two systems (Djekic et al., 2019; Notarnicola et al., 2017). In contrast, 6M have received primary data for the rescued bread flour manufacturing process from Rescued NZ.

System boundaries

The study employs a cradle-to-gate approach, excluding the downstream process; this reduces the scope of environmental hotspot analysis and the contribution from removed processes (see section 3.3.7). Although this is outside the scope, the study will treat these processes as scenarios (see section 5.7) to highlight any environmental hotspots. In this study, it is assumed that the downstream processes are similar for rescued bread flour and wheat flour (e.g., both products will be consumed or discarded).

2.9 Data Reliability and Coverage

The processes for both flours are identified, and all the resources required to produce each process are carefully selected to represent reality as closely as possible. In LCA, this qualitative check is referred to as the representativeness of the data, which involves three dimensions: the country of origin, the timeframe considered for the processes, and the technology used for the processes. The datasets might not perfectly represent the processes, so assumptions are made to select the most accurate data for the calculations.

The timeframe

The study sets the timeframe of the processes based on 2020-2025 values whenever possible; however, some data from the inventory dataset may be older than this. The milling of wheat into wheat flour and the production of rescued bread flour are assumed to occur in 2025.

The country of origin and technology used

Table 4 provide the information for the two products, to show what representativeness is considered. For transparency and potential reuse of the data, all data are logged into the LCI_metadata sheet with the assumptions made:

Table 4 - Specificity of data

Stages	Country of origin		Technology used	
	Rescued bread flour	**Wheat flour**	***Rescued bread flour***	**Wheat flour**
Raw material	Auckland, NZ	Australia	Industrial produced bread*)	Current and commonly used in the industry
Transport/ fuel	Auckland, NZ	Australia and NZ	100% Diesel truck 32 tonnes	Train, cargo ship, 32 tonnes diesel trucks
Production and machines	NZ	Australia and NZ	FDS And current and commonly used in	Current and commonly used in the industry
Energy use	NZ North Island Mix grid		Electricity mix and gas mix	Australian mix average

* Note: The industrial bread produced and received by Rescued NZ is unknown, the composition varies. All other ingredients required for commercial bread production are avoided, including salt, yeast, preservatives (if any), other flavoured components (if any), and sugar. Water is included in the surplus bread ingredient.

2.10 Preparing the Basis for the Impact Assessment

There are two approaches in the LCA final phase of interpreting the results of impact categories: (1) midpoint characterisation factors, within the cause-effect pathway (advanced technical interpretation of the impact scores) and (2) endpoint characterisation factors, at the end of the pathway (i.e., the effect), which reflect damage to three areas of protection (human health, ecosystem quality, and resource scarcity). Midpoint results provide a technical breakdown of multiple environmental issues, offering an in-depth representation of the potential environmental impacts associated with a process. In contrast, endpoint is easier to understand, as they provide a simpler justification of the impact category score by combining all the midpoint categories into three environmental areas only. The study will present the results with a combination of both midpoint and endpoint characterisation factors to present the most relevant interpretation for the results.

The study utilises openLCA software and the ecoinvent database for modelling the LCI phase, and ReCiPe 2016 for the LCIA phase. Figure 4 represent the impact categories from ReCiPe 2016 (Huijbregts et al., 2017, p.17). There are 17 impact categories in ReCiPe 2016; among these, five key indicators were selected for discussion. The selection was made according to the following criteria:

- Alignment with client priorities and the defined goal and scope of this study.
- Relevance to the agricultural and food production sectors, where these indicators are commonly regarded as the most representative environmental pressures.
- Direct linkages to the three endpoint damage categories (Human Health, Natural Environment, and Natural Resources) under the ReCiPe 2016 framework.

The five selected midpoint indicators are: Global warming, Fossil resource scarcity, Land use, Freshwater ecotoxicity, and Fine particulate matter formation. For clarity and better communication with the client, these are respectively renamed as:

- Carbon footprint
- Fossil fuel use
- Land use
- Freshwater toxicity
- Air pollution (PM2.5)

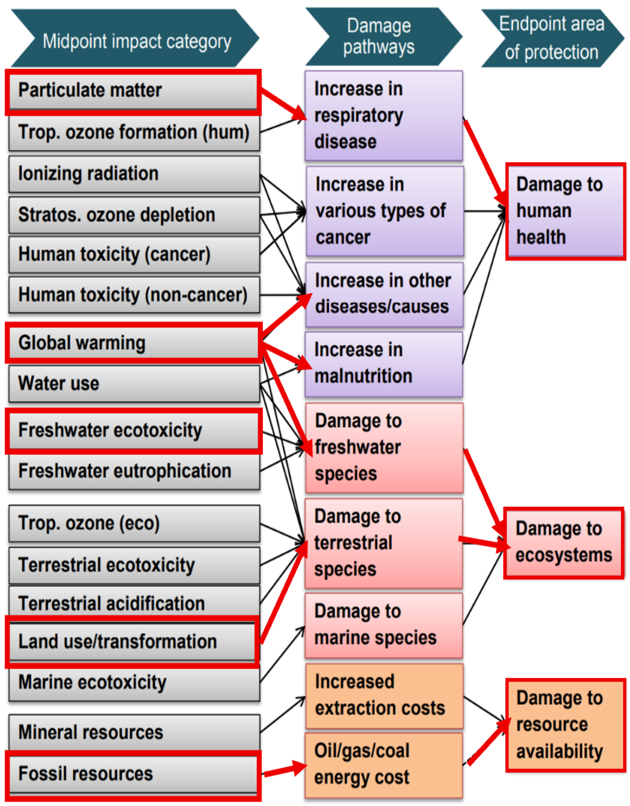

Figure 4 - Impact categories: Midpoint and endpoint (source: Huijbregts et al., 2017)

2.11 Special Requirements for System Comparisons

This report is not intended for public disclosure; however, the interpretation section of the report may include comparative assertions between the rescued bread flour and conventional wheat flour.

A critical review will not be conducted for this study, despite its comparative nature, as the results are not intended for public publication.

3. LCI

3.1 LCI Model at System Level

The LCI model at system model for the wheat flour in illustrated in Figure 5 and the LCI model for the rescued bread flour is illustrated in Figure 6. The elementary flows entering the system are resources and energy, the resources exiting the system are emissions. Those elementary flows are data quantified for the LCI modelling; they will determine the final impact score for each process. This representation is further explained in detail in section 4.3 System modelling.

In both LCI models, the flour waste such as dust and wheat husks are excluded from the study because their weight is insignificant (< 1% of total mass input). The wheat flour production process is complex (see Table 2), and all the processes are aggregated into one process in the ecoinvent 3.8 database, called "wheat grain processing".

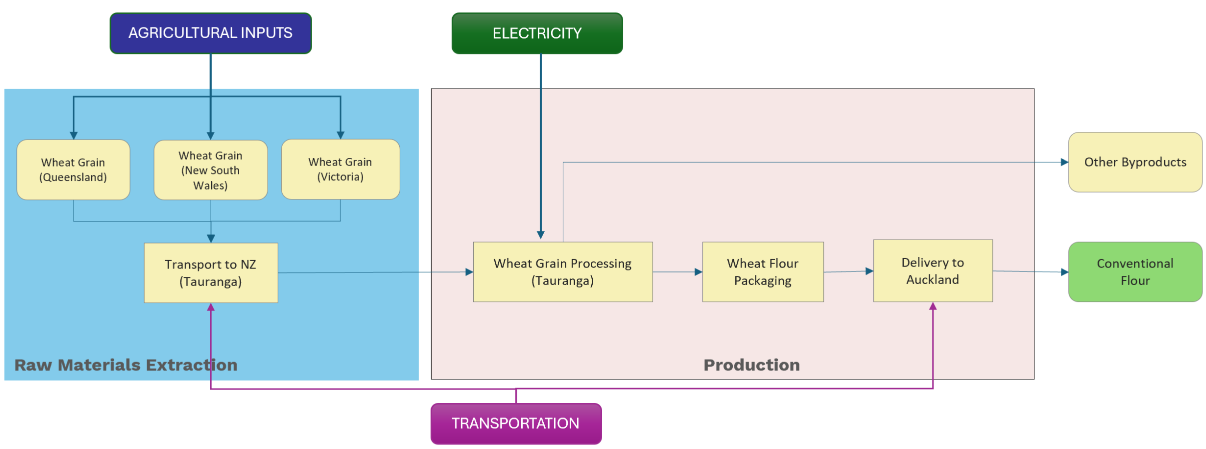

Figure 6 - Wheat flour LCI modelling

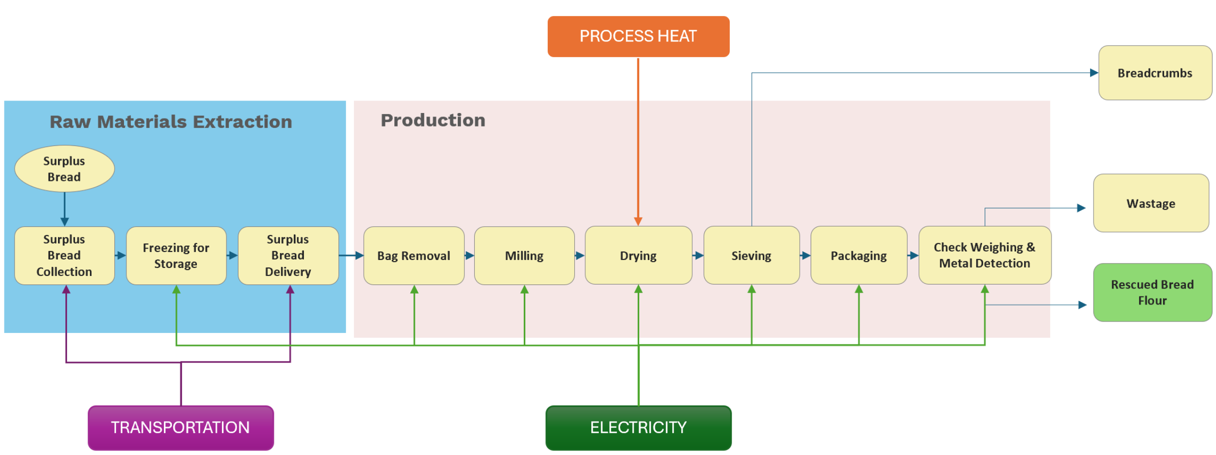

Figure 5 - Rescued bread flour LCI modelling

3.2 Data Collection

3.2.1 Primary Data

6M have received data on separate occasions from Diane Stanbara and Royce Bold on 25th August 2025:

- In-person presentation with Diane, followed by a Q&A session at the University of Auckland
- 22nd September 2025: Online Q&A with Royce, at the University of Auckland
- Two Q&A sheets from Diane where all questions were asked of different groups following the interviews with the clients.

The data received from the clients is considered as primary data, and all the other data from this study are selected according to their relevance and any assumptions made are shared throughout the report.

3.2.2 Secondary Data

Primary data is considered a high-quality data, as it is usually observed, measured and quantified directly on-site by the client. Any other data, or secondary data, is therefore less qualitative. For this study, all secondary data are derived from ecoinvent 3.8 database (medium quality) and literature publicly available (low quality). For instance, all the data for wheat flour are provided by ecoinvent 3.8 with information collected through literature to refine the model.

3.2.3 Documentation for all metadata and LCI results

All the data for rescued bread flour are logged into a LCI framework presented in Appendix A. It gathers all the information and details such as quality of data, value, unit, source, access, provider (if any), and most importantly, the assumptions, calculations and any relevant comments to justify the selection of data. It is important to store the data for transparency, tracking and for potential reuse of the data for a sensitivity analysis or another comparative LCA study.

Once the software has calculated the impact score, the LCI results are documented as a list of quantified elementary flows (resources, energy and emissions) is produced.

3.3 Constructing the LCI Model

Table 5 details the data inputs for the rescued bread flour with the elementary flows quantified.

Table 5 - LCI data for rescued bread flour

ID	Parameter input	Value	Unit	Note
Raw material extraction				
	Surplus bread	1644	kg	The soft plastic is removed from the LCI, following the cut-off threshold criteria for the study, as it mass is< 1% of the total mass of the bread and packaging.
	Soft plastic bag	0	kg	
B1	*Bread collection transport*	20	km	The calculation for the collection mileage is based on an average of all Woolworths supermarkets in the Auckland region.
B2	**Freezer Storage**			
	Electricity	77	kWh	Decrease temperature from 20ºC to –15ºC
B3	**Transport**			
	Delivery	12	km	The distance between Kiwiharvest in East Tamaki and Foodbowl in Mangere
Production processes				
B4	**Debagging**			
	Input bread weight	1644	kg	
	Electricity	34	kWh	Conveyor belt electricity
B5	**Dehydration FSD**			
	Combustion Process heat	4493	MJ/kg	
	Electricity	70	kWh	
Output				
	Output water vapour to air	457	kg	27.5% water waste given by Rescued NZ
	Output bread pellets	1206	kg	72.5% bread outputs
B6	**Fine milling**			
	Electricity	43	kWh	
B7	**Sieving**			
	Electricity	11	kWh	
Output				
	Output *rescued bread flour* (85%)	1025	kg	
	Output breadcrumbs (15%)	181	kg	
B8	**Packing**			
	Rescued bread flour	1033	kg	
	Paper bag	7	kg	

	Electricity	25	kWh	
B9	**Quality control**			
	Electricity	11	kWh	X-ray metal detectors and scale to weight final product
Output				
	Rejected rescued bread flour (2.5%)	25.8	kg	Information given by Rescued NZ
	Rescued bread flour, packed	1007	kg	*Rescued bread flour* with packaging

Table 6 details the data input for the wheat flour with the elementary flows quantified.

Table 6 - LCI data for wheat flour

ID	Parameter input	Value	Unit	Note
Raw material extraction				
C1	*Wheat grain*	1000	kg	
	Import to NZ	1504	km	
Production processes				
C2	**Wheat grain processing**			
	Water		kg	Tap water for cleaning
	Electricity		kWh	
C3	**Packaging & Delivery**			C3 – Aggregated in one process on OpenLCA
	Paper sack	7	kg	
	Electricity		kWh	
	Transport to Auckland		km	
Output				
	Wheat flour, packed	1007	kg	

3.4 Calculated LCI Results

The totality of LCI results from the ***rescued bread flour***, the wheat flour and the comparison in between the two products are presented in Appendix B.

3.5 Reporting

The study has selected five impact categories that are the most relevant to answer the goal and scope. Table 7 gives a definition of the environmental impacts related to each selected category as follows.

Table 7 - Definition of midpoint category

Midpoint Impact category	Environmental impact
Carbon footprint	Measures the total emissions of GHG, expressed as kilograms of CO_2-equivalent (kg CO_2-eq). This indicator measures the extent to which a process contributes to global climate change through energy use, fuel combustion, and other emission sources.
Land Use	Indicates the amount of agricultural or forestry land occupied throughout the product's life cycle, measured in square meters per year (m^2·year crop-eq). This category captures the competition for land between food production and other ecosystem services.
Fossil fuel use	Represents the depletion of non-renewable fossil resources such as coal, oil, and natural gas, expressed as kilograms of oil equivalent (kg oil-eq). It reflects energy dependence and the potential for long-term resource scarcity.
Freshwater Ecotoxicity	Measures the potential harm to freshwater organisms (e.g., fish, algae, and invertebrates) caused by chemical emissions such as fertilisers, pesticides, or industrial effluents. It is expressed as kilograms of 1,4-dichlorobenzene equivalents (kg 1,4-DCB-eq).
Air pollution PM2.5	Represents the formation of fine particulate matter (PM2.5) that can affect air quality and human respiratory health, expressed in kilograms of PM2.5-equivalent. It captures the impact of air pollution from combustion and energy-related processes.

Note: The choice of the five selected impact categories is explained in Section 2.10 of the study's scope.

3.5.1 LCI results

First, all the data from the LCI phase are calculated and are represented into their characterisation factor, or unit. It represents the quantity of an elementary flow for an impact category (e.g., carbon footprint of freezing process is equal to 12 kg CO_2 eq). To simplify the interpretation of the results, the LCA practitioner defines the process or aggregation of processes to calculate, and the software will automatically calculate the results per impact category.

Table 8 summarises the total impact scores of the two flours for each selected impact category. This representation helps to understand the primary differences between the two products in terms of potential environmental harm.

The assumptions for the study are presented throughout the report, the following Table 9 summarises the main assumptions for both flours:

Table 8 - LCI results comparison summary

Impact categories	Unit	*Rescued bread flour*	Wheat flour	Change % (Wheat flour to RBF)
Carbon footprint	kg CO_2 eq	405	1787	↓77%
Fossil fuel use	kg PM2.5 eq	106	144	↓26%
Land use	kg oil eq	16	6001	↓100%
Freshwater toxicity	kg 1,4-DCE	12	26	↓54%
Air pollution (PM2.5)	m^2a crop eq	0.26	3.67	↓93%

Table 9 - Summary of assumptions

Category	Rescued bread flour	Conventional wheat flour
Raw material extraction	The surplus bread composition of bread is unknown, and the study uses "surplus bread" as a custom flow.	The wheat grains production relies solely on ecoinvent 3.8 database.
Transport	KiwiHarvest has a mix fleet of electric vehicle and diesel refrigerated truck, which can be used interchangeably. As it is impossible to know which truck is used, the study considers the use of 100% diesel trucks as the baseline. Refrigerated trucks are modelled in LCI by ecoinvent 3.8 The LCI modelling uses EURO 5 trucks	The wheat grains are transported from farm to port using a mix of rail and road transports. The importation from Australia is done by bulk carrier ship to NZ. The transport from Auckland to Tauranga The LCI modelling uses EURO5 trucks
Production processes	Primary data is used for the energy consumption of the whole production line except of the freezing stage which is modelled based on ecoinvent 3.8. The upscale scenario does not include a freezing process, although the study considers this phase as current practice.	The milling process takes place in Tauranga, NZ.
Packaging	The soft plastic of surplus bread is out of the system boundaries Rescued NZ provided information about their packaging supplier (Oji Solutions), although no further information of composition and size where given; thus the final packaging is a 15kg paper sack using ecoinvent 3.8 dataset.	The wheat grains are transported in bulk, in containers until they reach the milling factory in Tauranga, no packaging is used The wheat flour packaging for storage is the same as rescued bread flour (15kg paper sack)
Storage		The final products are transported to Auckland following food safe transport practices and stored in an ambient room temperature in Auckland. For the study, both flours are assumed to be stored in the same conditions.

4. LCIA

The results from the LCIA phase were assessed using the software. The previous section presented the total result of the comparison between the two flours, which serves as the foundation for organising the data and providing an evaluation and interpretation of the results. The study's goal and scope define the intended application for these results:

- The comparison of potential environmental impact between the two flours,
- The environmental hotspots in Rescued NZ's processes, and
- The environmental performance of the rescued bread flour with concrete and science-based data.

The following section breaks down the data to answer the goal and scope of the study.

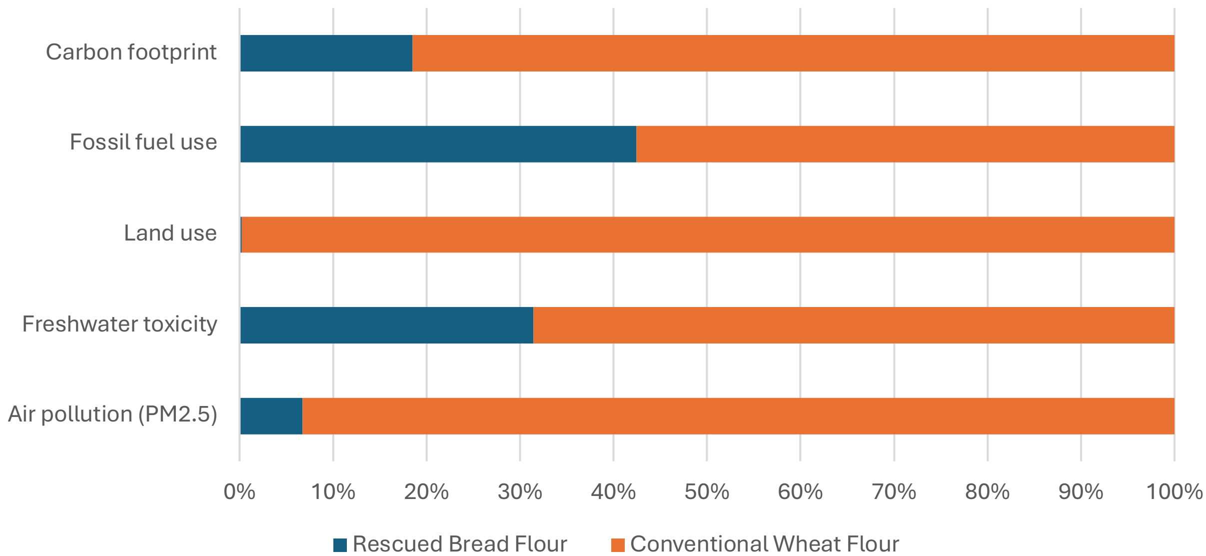

Figure 7 - Representation of LCIA results in bar graph

5. Interpretation

5.1 Product comparison: Midpoint impact contribution

The interpretation phase identifies the important elements behind the results in Table 8 by examining each production process separately. Figure 7 presents the results in a bar chart to clearly illustrate the variation in numbers. Based on these results, the study can confirm that the rescued bread flour significantly outperforms wheat flour in all five categories:

Carbon footprint

The production of 1 tonne of rescued bread flour produces only around 23% of the carbon emissions compared to the production of 1 tonne of wheat flour. This improvement primarily results from reducing emissions associated with wheat grain cultivation, harvesting, and long-distance transportation, which are the most carbon-intensive stages in conventional production.

Fossil fuel use

The rescued bread flour is approximately 26% lower in fossil fuel use compared to wheat flour. This outcome arises from the fact that Rescued NZ's operations rely solely on collection and processing, thereby avoiding the heavy fuel consumption associated with field operations and transportation.

Land use

The land use is the most significant variation between the two flours, showing a difference of nearly 100%. This is attributed to the absence of farming processes in the production of the rescued bread flour.

Freshwater toxicity

The result for the rescued bread flour in freshwater toxicity shows a reduction of approximately 54% compared to the wheat flour. This is primarily a consequence of the intensive farming processes for wheat grains (i.e., soil preparation, crop growth, and harvesting), with the use of fertilisers and pesticides. Chemical runoff is a common waste from agricultural processes and significantly impacts freshwater streams and their biodiversity. While the does not discharge any chemicals from its production processes, the 30% contribution to freshwater ecotoxicity can be attributed to the other elements from the production line.

Air pollution (PM2.5)

Air pollution is another impact that shows significant variance between the two flours. Air pollution is a consequence of fine particles emitted into the air by the use of transport engines (e.g., diesel-powered fleets, cargo ships, and trains), as well as the combustion of machinery. Hence, the 93% reduction from the rescued bread flour is attributed to the absence of agricultural machinery use, reduced transportation and, as a result, reduced combustion activities.

This is the first interpretation that answers two of the three intended applications: the comparative results of potential environmental impacts from the two flours and the overall environmental performance (midpoint categories) of the rescued bread flour.

5.2 Product performance: Endpoint impact contribution

The comparison results between the two flours clearly demonstrate the superiority of the rescued bread flour over the wheat flour. The endpoint impact contribution also assesses the comparison between the two flours within the three areas of protection: Damage to human health, damage to ecosystems and damage to resource availability (Huijbregts et al., 2017). They have been renamed as Human health, Natural environment, and Natural resources to simplify their understanding. To understand the link between midpoint and endpoint categories. Figure 4 (section 2.10) illustrates the relationship between the two categories.

The definition of endpoint categories is presented in Table 10:

Table 10 - Definition of endpoint category

Endpoint Impact category	Environmental impact
Human Health	Measures total damage to human health and is expressed as disability-adjusted life years (DALY). This endpoint converts exposure to stressors such as air pollution, ozone, carbon footprint and toxic releases into years of healthy life lost through mortality and morbidity along the life cycle.
Natural Environment	Measures damage to ecosystems as loss of native species over time, expressed as species year. It integrates the fraction of species that potentially disappear across the affected area and duration. This category captures biodiversity loss driven by land occupation and transformation, ecotoxic emissions and other pressures that reduce species persistence.
Natural Resources	Measures the decline in future resource availability as additional extraction cost, expressed as USD2013. It reflects the surplus cost required to obtain minerals and fossil resources as high-quality deposits are depleted. This category captures long-term pressure on resource stocks from energy use and material inputs throughout the system, linking current consumption to future accessibility.

The study has selected midpoint categories to reflect the potential impact of each area of protection, which simplifies the interpretation of the midpoint categories. Moreover, it clearly identifies the relative assumptions of environmental harm associated with intensive farming processes. The results are readily interpretable for the general public.

Table 11 summarises the results for the two flours, while Figure 8 provides a visual representation of those findings.

Table 11 - Summary of endpoint category scores

Impact category	Unit	Rescued Bread Flour	Wheat Flour	Change
Human health	DALY	6.14E-04	4.16E-03	↓ 85%
Natural resources	USD2013	3.66E+01	5.00E+01	↓ 27%
Natural environment	species.yr	1.51E-06	6.31E-05	↓ 98%

As previously observed, at the endpoint level, the rescued bread flour shows significant reductions across all three categories:

Human Health

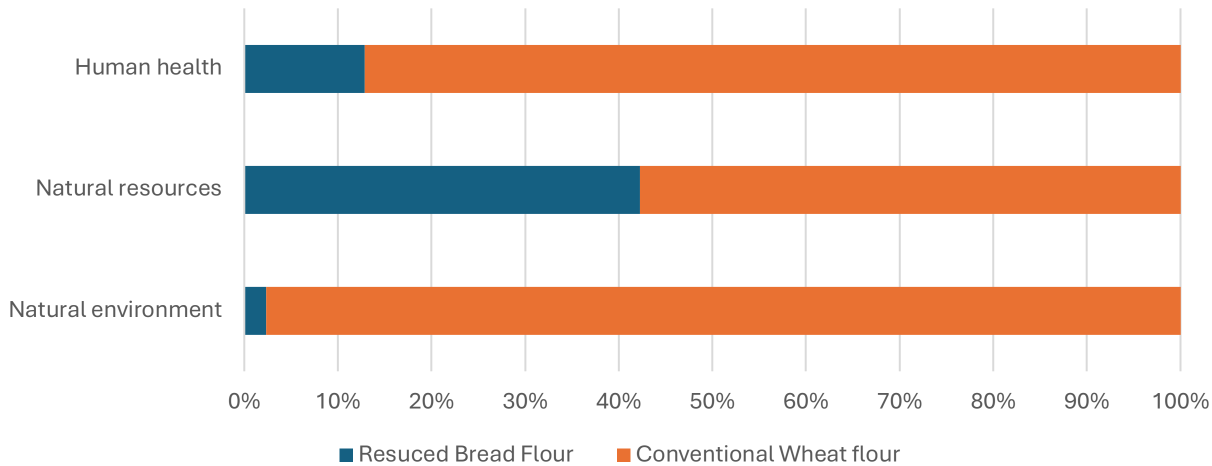

Figure 8 - Representation of summary of endpoint category scores in bar graph

A substantial improvement in human health impact stems from lower impacts on air pollution and greenhouse gas emissions (carbon footprint). The rescued bread flour performs better than the wheat flour for the human health indicator. A substantial improvement in human health impact stems from lower impacts on air pollution and greenhouse gas emissions (carbon footprint), both of which drive years of healthy life lost. By avoiding field preparation, crop growth and harvesting, the system removes diesel use, fertiliser emissions and dust that normally add to respiratory impacts. Shorter and lighter logistics further limit combustion-related pollutants. Remaining contributions mainly come from process heat in drying and electricity demand. Although these loads are significantly smaller than for the farming process, which reflects the overall variance in the two flours.

Natural Resources

A 27% decrease in natural resource depletion is driven by a reduction in the consumption of fossil energy and raw materials per tonne of flour delivered. The rescued bread flour system eliminates energy from cultivation and grain processing, which lowers extraction pressure on fuels and minerals. Additionally, transport demand is reduced to collection and short transfer to the production site, which reduce diesel consumption. The contribution from the drying process and paper packaging is still lower than in the wheat flour route, hence, the endpoint indicates a meaningful decrease in resource damage.

Natural Environment

A notable reduction in damage to ecosystems is due to the decreased land occupation and freshwater toxicity which protects habitat quality. Lower releases of toxic substances from fertilisers and pesticides reduce stress to freshwater organisms, and fewer combustion emissions cut nitrogen and sulphur that can alter ecosystems. Drying and packaging's ecological footprints are outweighed by the avoided farming processes. The natural environment endpoint confirms a pronounced reduction in potential species loss across terrestrial and aquatic systems as compared to wheat flour.

The interpretation under the endpoint category strengthens the comparison between the two products and optimises the product performance comprehension.

5.3 Environmental hotspots

The third objective of the study focuses on identifying the environmental hotspot from Rescued NZ's production line. The results derived from the contributions of each process to the selected impact categories are presented in the Table 12. These results give a quantitative understanding of the processes' weight within the categories, although a qualitative interpretation is essential to evaluate which element of the processes is responsible for the impact. Figure 9 summarises the results in bar charts to facilitate their reading.

Table 12 - Rescued bread flour's processes contribution to environmental impact

Impact Category	Carbon footprint	Fossil fuel use	Land use	Freshwater toxicity	Air pollution (PM2.5)
Unit	kg CO_2 eq	kg oil eq	m^2a crop eq	kg 1,4-DCB	kg PM2.5 eq
Collection	19.40	6.01	0.41	0.85	0.02
Freezing	4.02	1.31	0.02	0.27	0.01
Delivery	14.17	4.39	0.30	0.62	0.01
Bag Removal	3.47	1.12	0.02	0.96	0.01
Milling	4.50	1.46	0.02	1.24	0.01
Drying	344.62	87.20	1.45	5.87	0.17
Sieving	1.19	0.38	0.01	0.33	0.00
Packaging	12.61	3.83	13.59	1.28	0.03
Check	1.36	0.44	0.01	0.38	0.00
Rescued Bread Flour	405.33	106.13	15.81	11.79	0.26

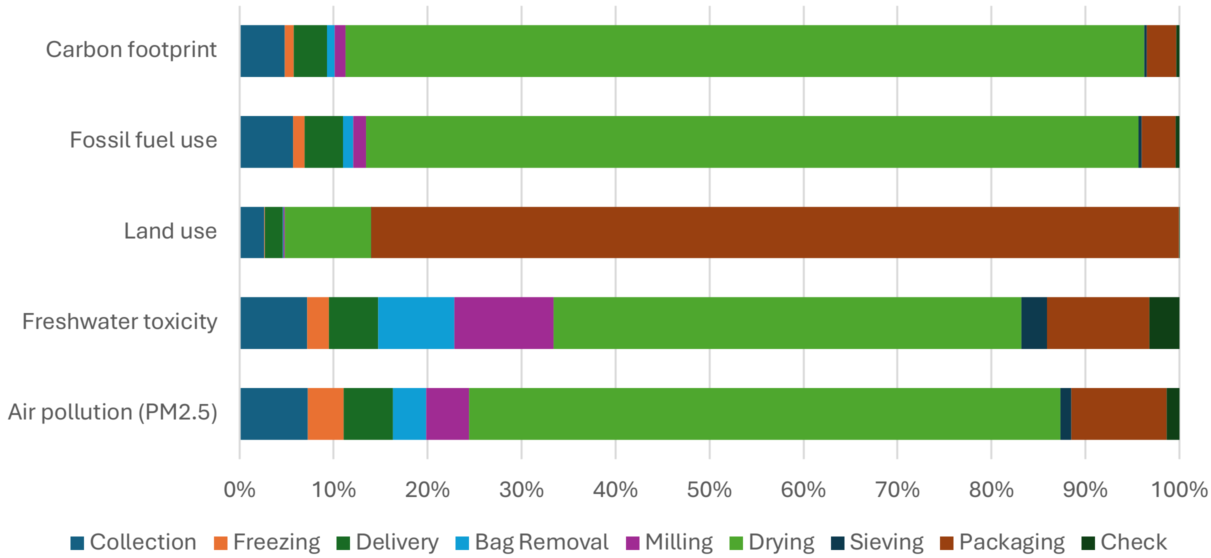

Figure 9 - Representation of rescued bread flour's processes contribution to environmental impact

5.3.1 Hotspot #1: Drying methods

The FSD process is the single largest contributor to all five indicators. It is particularly notable in terms of carbon footprint and fossil fuel use, accounting for 85% and 82% of the totals, respectively. It is also the largest contributor to freshwater toxicity (49%) and air pollution (PM2.5) (63%). These shares result from the evaporation of water from the bread; this mechanism is an energy-intensive step that utilises both gas and electricity. It represents the main source of both greenhouse gas emissions and non-renewable energy consumption; moreover, it drives substantial upstream chemical and combustion-related emissions that affect water toxicity and particulate formation.

5.3.2 Hotspot #2: Packaging

Packaging has a significant impact on land use, accounting for 86% of the total impact indicator. This large share reflects the upstream processes where land requirements embedded in the paper-based bulk bags used for packaging (i.e., the agricultural and forestry land area required to produce the raw materials). Although the packaging mass per tonne is relatively small, the lifecycle land occupation associated with paper production makes it the primary hotspot for land use.

5.3.3 Exploring other factors

Freshwater toxicity and air pollution, as measured by PM2.5, are both indicators that are relatively distributed across multiple processes, in contrast to carbon footprint or fossil fuel use. The underlying cause is explained by the collection, debagging, milling, and packaging processes, which each contribute to some degree to the environmental degradation. The transport sector (collection and delivery) contributes significantly to three indicators, accounting for 12% of freshwater toxicity, 13% of air pollution, and 10% of fossil use; this is due to the use of diesel fuel and associated intensive production and combustion waste. Additionally, the milling process contributes to 11% of freshwater ecotoxicity and 5% of air pollution, primarily due to the release of bread dust waste into the air. Although those last results derive from the ecoinvent 3.8 database, which considers milling from industrial procedures, rather than from a small-scale commercial kitchen or laboratory. Therefore, those results might be underestimated in reality.

Other stages, such as sieving and final check, are minor contributors across all five indicators (each typically accounting for less than 1–3% in the categories as shown on the Figure 9). Those processes rely on electricity consumption.

In practice, this means that, in addition to the hotspots, drying and packaging, other secondary process points (i.e., transport and milling) also deserve attention. The interpretation first focuses on the main contributors to the impact; nonetheless, other incremental improvements on minor contributors can reduce the overall performance of the rescued bread flour.

5.4 Identification of Significant Issues

From this point forward, the study presents the documentation for the overall quality of the chosen methodology and the selected data. To maintain transparency, the precision of the data is provided to the reader; therefore, an uncertainty analysis is also included. The identification of significant issues summarised in Table 13 provides two things:

- For the practitioner, a checklist to refer to while collecting data and ensuring alignment with the goal and scope of the study
- For the reader, an overview of the potential issues highlighted during the study.

Table 13 - Identification of significant issues

Items Checked	Results
Goal and scope	
Methodological approach	Both the ***rescued bread flour*** and wheat flour systems employed the same LCIA method (ReCiPe 2016 midpoint) and identical impact categories, enabling a valid comparison. The functional unit was consistently defined as 1 tonne of flour used for baking, ensuring an equivalent basis for comparison despite the different upstream production pathways. System boundaries for both were consistently established as cradle-to-gate, encompassing raw material extraction, processing, and packaging, but excluding the use and end-of-life phases.
Allocation procedures	Both systems have multi-output processes, and allocation methods were applied consistently. For wheat milling, the co-products of flour are excluded. The breadcrumbs from the ***rescued bread flour*** processes were handled under mass allocation.
LCI	
Data quality	The information for ***rescued bread flour*** were mostly available from **Rescued NZ**, while for wheat flour were obtained from the average industry process in the ecoinvent 3.8 database. The secondary data were also sourced consistently from the ecoinvent 3.8 and the literature for both systems. The country of origin and technology used were documented transparently. The timeframe was maintained by utilizing data from equivalent years where possible.
Assumptions	All assumptions for the processes, inputs, and outputs were transparently documented and justified by literature or industry studies.
LCIA and interpretation	
Impact categories	Five impact categories were chosen and applied consistently for both products, considered the client needs, impact categories related to agricultural emission burden from wheat flour, and encompassing all three endpoint impact categories.

5.5 Sensitivity and Uncertainties analysis

In LCA, uncertainties exist in all phases, whether in methodology choices, mistakes or simply parameters. This is the reason all assumptions are disclosed, to manage the uncertainties in interpreting the results. Identifying and managing uncertainties is an important element of an LCA study; it improves the precision of the study through iteration (repetition of the calculation of the results) when fine-tuning the parameters. A sensitivity analysis allows the practitioner

Indicators & Scores

	1	2	3	4	5
Reliability	Verified data based on measurements	Verified data partly based on assumptions or non-verified data based on measurements	Non-verified data partly based on qualified estimates	Qualified estimate (e.g. by industrial expert)	Non-qualified estimates
Completeness	Representative data from all sites relevant for the market considered, over and adequate period to even out normal fluctuations	Representative data from > 50% of the sites relevant for the market considered, over an adequate period to even out normal fluctuations	Representative data from only some sites (<< 50%) relevant for the market considered or > 50% of sites but from shorter periods	Representative data from only one site relevant for the market considered or some sites but from shorter periods	Representativeness unknown or data from a small number of sites and from shorter periods

B8 - Packaging				Pedigree Matrix						
	Amount	Unit	Source	Reliability	Completeness	Temporal Correlation	Geographical Correlation	Further technological correlation	Data quality entry	Database provider
Primary Output										
Rescued Bread Flour, packaged | Reference Flow | Rescued Case	1.007364	kg	Process Output						B9	
Inputs										
electricity, low voltage	0.015327813	kWh	ecoinvent 3.8	2	1	2	1	3	(2; 1; 2; 1; 3)	market for electricity, low voltage | electricity, low voltage | Cutoff, U - NZ
paper sack	0.007364	kg	ecoinvent 3.8	4	2	5	2	1	(4; 2; 5; 2; 1)	market for paper sack | paper sack | Cutoff, U
Rescued Bread Flour	1	kg	Process Input						B08	B08 - Sieving, separation of flour and breadcrumbs

Figure 8 - Pedigree matrix on openLCA and an example of data quality uncertainty.

to test the sensitivity of the parameters, for instance. In the case of the paper sack for rescued bread flour, if it becomes slightly lighter or heavier, how much change would be observed in the results? If the difference is negligible, it can be concluded that this parameter is not sensitive. Conversely, if a slight change leads to a significant variance, then it means two things: the paper sack mass is a sensitive element, and it is worth understanding what causes the sensitivity and the associated decrease or increase of environmental harm.

5.5.1 Uncertainty analysis documentation

The study identifies the uncertainty of the database for all LCI data across each phase. Each data point is scored to assess its quality in terms of five indicators: completeness, reliability, and specificity (country of origin, technology, and time frame). This is called the Pedigree

Matrix method. Figure 8 also illustrates part of the scoring system for data quality in openLCA and below, an example of uncertainty analysis for packaging process. The scores range from 1 to 5, "1" being the most certain and "5" the least certain. The final pedigree matrix is presented in Appendix C.

From the analysis of the Pedigree Matrix, rescued bread flour has lower score than the wheat flour as it relies mainly on primary data.

5.5.2 Sensitivity analysis

Every parameter can be subject to a sensitivity analysis; the more, the better, as it refines the accuracy of the results and determines which parameters are more or less sensitive to the model. This study presents a sensitivity analysis for several parameters. The detailed results are presented in Appendix D.

Functional Unit

The functional unit has been selected for a sensitivity analysis as it defines the core of the study, it is a sensitive parameter to the mass variance. The study examines the changes in production mass when it increases or decreases by testing the sensitivity for different functional unit mass: 0.5 tonnes, 1 tonne, and 1.5 tonnes. The results show that the functional unit is not a sensitive parameter for those impact categories. Although the carbon footprint is affected by the changes, it is evident that the rescued bread flour (functional unit = 1.5 tonnes) produces just a third of the carbon footprint of wheat flour (functional unit = 1 tonne).The study can conclude that the rescued bread flour performs better in terms of carbon emissions than the wheat flour, even with an increased production capacity.

Propane

Propane is one of the most common gases used in Liquefied Petroleum Gas (LPG). The modelling for the study defined propane in mass (kg) as part of the heat production unit process. The produced heat is utilised for the FSD process.

The sensitivity analysis investigates the effects on the impact categories of using more or less propane (e.g., +50% and -50%) for the FSD, assuming more or less efficient gas consumption. The result shows that the input of propane has a significant impact on fossil fuel use, as opposed to the FU analysis. This result is attributed to the nature of propane production.

Electrical Efficiency

Electrical efficiency is related to the electrical load, measured in kilowatt-hours (kWh), required to operate the equipment. Electricity production relies on the remarkable properties of copper. Copper has superior electrical and thermal conductivity compared to other materials (ICA, 2021). Metal copper is among the most ubiquitous contaminants in freshwater aquatic environments; due to its widespread use, it greatly affects the freshwater biodiversity and ecosystem (e.g., fish, invertebrates, plants, and algae) at high concentrations (Ministry for

Environment, 2025). In the production line, the FSD equipment has the highest electrical load among all unit processes. The study examined the change in electricity efficiency by assessing a system that is 50% more efficient (e.g., use of more energy efficient machines) or 50% less efficient (e.g., equipment ageing or poor maintenance).

The result show that the electricity efficiency is sensitive particularly to freshwater toxicity. The other impact categories are not affected as much. Additionally, it is observed that the rescued bread flour performs better in freshwater toxicity consistently, even with a 50% decrease in efficiency. The study can conclude that the energy consumption uses for the rescued bread flour is not targeted as an environmental hotspot, despite the high energy use of the FSD. This is partly due to the minimal use of heavy equipment used by Rescued NZ.

5.5.3 Contribution Analysis

There are multiple parameters used in the LCI modelling for both flours, a contribution analysis helps estimating the uncertainty range of those parameters for each impact category. The study uses the Monte Carlo simulation that generates 1000 iterations, using a 90% confidence level. Detailed information on the statistical distribution for selected input variables can be found in Appendix E. The uncertainty analysis focuses on two principal indicators: carbon footprint and fossil fuel use. The uncertainty in this study arises mainly from:

- variability in input data from secondary databases (ecoinvent 3.8),
- assumptions in packaging materials and energy sources, and
- model simplifications such as mass allocation and average transport distances,

as these were accounted for in the Monte Carlo simulation to represent parameter uncertainty.

The analysis shows a strong model reliability for the rescued bread flour system. In contrast, the high variability of wheat flour highlights the uncertainty of the agricultural systems used in the LCI modelling. For both flours, the results are very close to the mean values obtained from Monte Carlo simulation. This means the model is well balanced and not dominated by outlier parameters. Furthermore, compared to wheat flour, rescued bread flour has lower model uncertainty and higher result reliability due to a lower coefficient of variation.

Lastly, the comparative advantage of rescued bread flour remains consistent across the Monte Carlo trials, confirming that waste valorisation significantly reduces environmental impact even when uncertainty is considered.

5.6 Completeness and Consistency check

The completeness check verifies that sufficient information from all LCA phases has been gathered to reach robust conclusions aligned with the study's goal and scope definition. All the information is presented under Section 5.4 as shown on Table 13.

The consistency check investigates whether assumptions, methods, and data have been applied uniformly throughout the LCA study in alignment with the stated goal and scope.

The study has documented the detailed completeness and consistency checklist presented in Appendix F.

5.7 Scenario Analysis

A scenario analysis serves as a valuable tool to generate more insights to improve the production processes. The study presented environmental hotspots in Section 5.3, which serves as a foundation for evaluating changes that can be implemented to reduce the impacts. Additionally, the study explores other scenarios that are relevant to potential improvements.

Scenario A: Freezing

In this scenario the freezing energy requirement is changed. It is either doubled or removed entirely. From the radar chart in Figure 9, it can be observed that Air Pollution (PM2.5) is most impacted by either doubling freezing time or removing the freezing process. Freezing are indirect but significant contributors to Air pollution through energy use. The source of energy that contributes the most to Air Pollution (PM2.5) is from lignite or coal.

From the bar chart in Figure 10, it can be inferred that different refrigeration scenarios have less than 5% impact on Air Pollution PM2.5. For comparison with Conventional Wheat Flour, Figure N indicates that all three freezing scenarios produce dramatically less, or roughly 13 times less, Air Pollution PM2.5 compared to Conventional Wheat Flour.

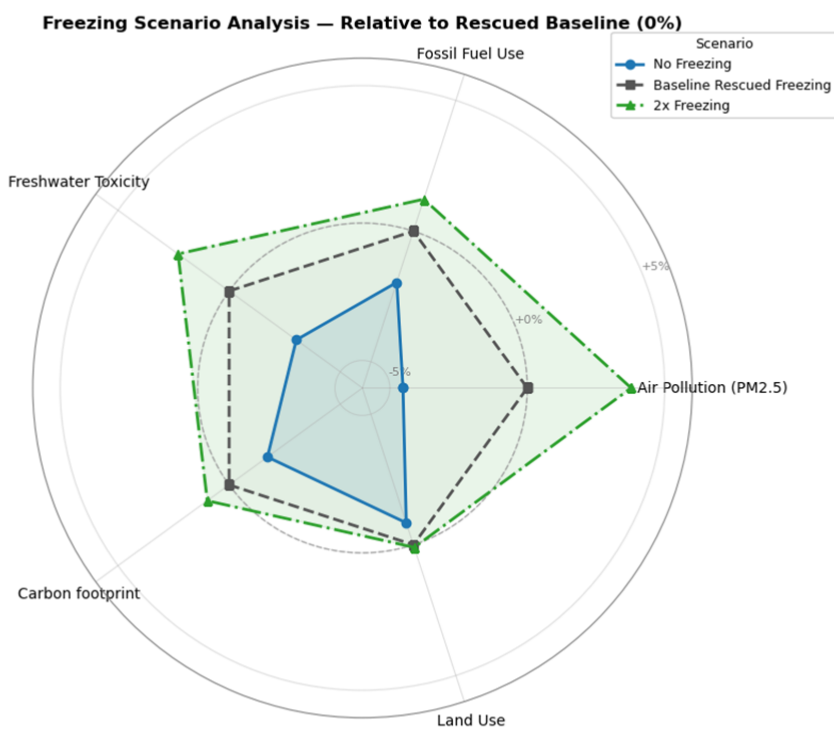

Figure 9 - Radar chart of freezing scenarios for rescued bread flour

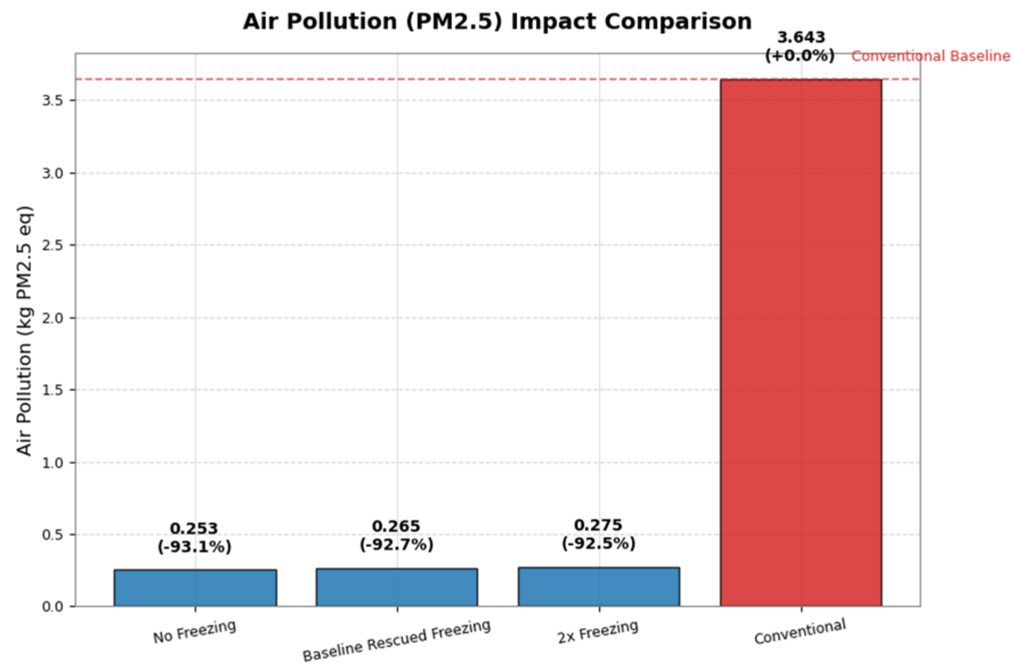

Figure 10 - Comparison air pollution impacts of rescued bread flour freezing scenarios and conventional flour.

Scenario B: Electrifying the drier

The scenario investigates the electrification of the FSD to eliminate the use of gas, which is responsible for almost 82% of the total impacts. The electrification of the FSD can be implemented by using an electric boiler to generate heat energy for the dryer. The results show a decrease of more than 50% in carbon footprint and 41% in fossil fuel use, which is attributed to the elimination of direct propane combustion and reduced upstream extraction of fossil resources. However, the results also reveal a notable trade-off in freshwater toxicity (73% increase), and air pollution (55% increase). The production and combustion of these fuels, as well as metal and chemical emissions from power plant operations, contribute significantly to aquatic toxicity and particulate matter formation.

The study finds that electrifying the drying process is the most efficient method to immediately drastically decrease emissions.

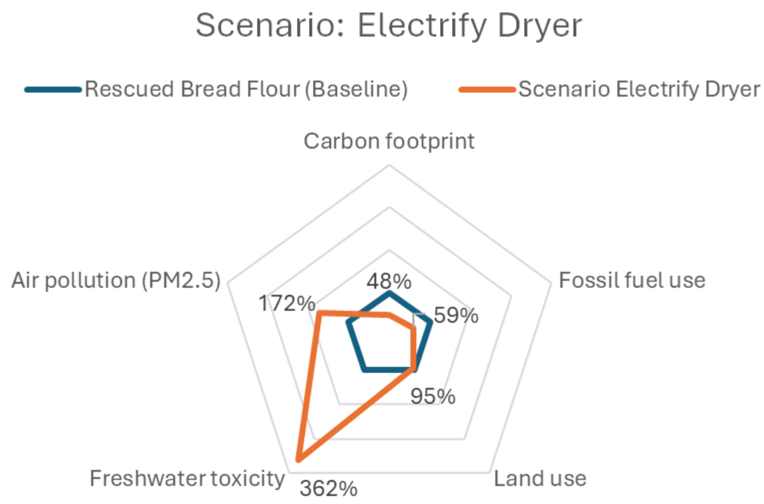

Figure 11 - Baseline drying process vs electric drying

Scenario C: Reusing the paper sack

The paper sack is the second contributor to environmental degradation from land use (intensive forestry operations to produce paper), accounting for 87% of the total for the category. Ideally, a paper sack can be reused as many as 43 times, according to a Danish study (Danish Environmental Protection Agency, 2018) that investigated the issue. To simplify the scenario model, the study considers a reuse of 10 times.

The results show a reduction of approximately 78% in land use; moreover, all other impact indicators show a decrease at various levels, reflecting lower resource extraction and energy requirements in manufacturing.

Reusing paper sacks requires some logistics and collaboration from the end-user to maintain the bags, as well as for Rescued NZ to organise the collection, tracking, and efficiency of the reuse system, which they communicate to their customers.

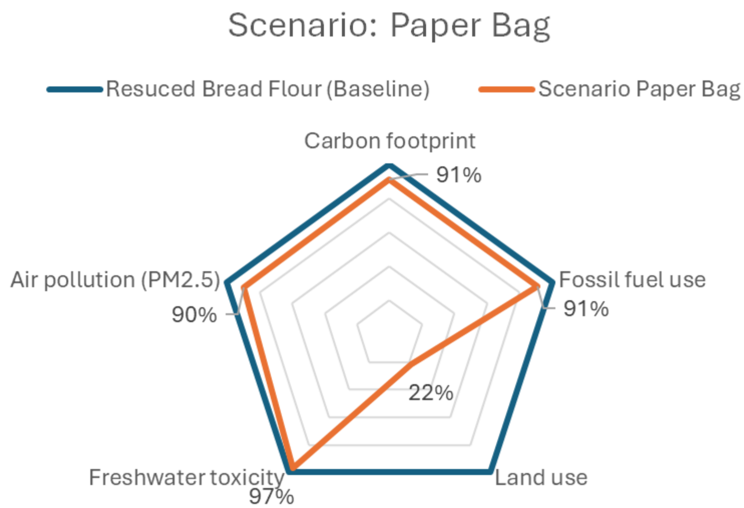

Figure 12 - Impacts of baseline and re-using packaging

Scenario D: Regulation changes in transportation

The study anticipates a change in regulation for transportation in 2026 as a national initiative to reduce carbon emissions. The change follows Australian law to implement the EURO6 standard vehicle; based on (Ministry of Transport, 2023), NZ is planning to shift from EURO5 to EURO6 incrementally from 2024 until 2028. Furthermore, tests have shown that EURO6 standards could reduce the exhaust, such as air pollution (PM2.5), by half (ICCT, 2016). The current modelling used EURO5 for all trucks; hence, this scenario impacts both the wheat flour and rescued bread flour impact scores.

The result shows that Rescued and Conventional wheat flour experience a reduction in air pollution PM2.5 impact after the EURO 6 trucking standards are implemented, by 5.7% and 3.3%, respectively. This reduction is unexpectedly small, although this can be explained from the fact that the study considers the impact of air pollution PM2.5 on all unit processes, not just transportation. A significant amount of air pollution PM2.5 emissions flows are actually observed within the drying and milling processes (from emission of volatile compounds).

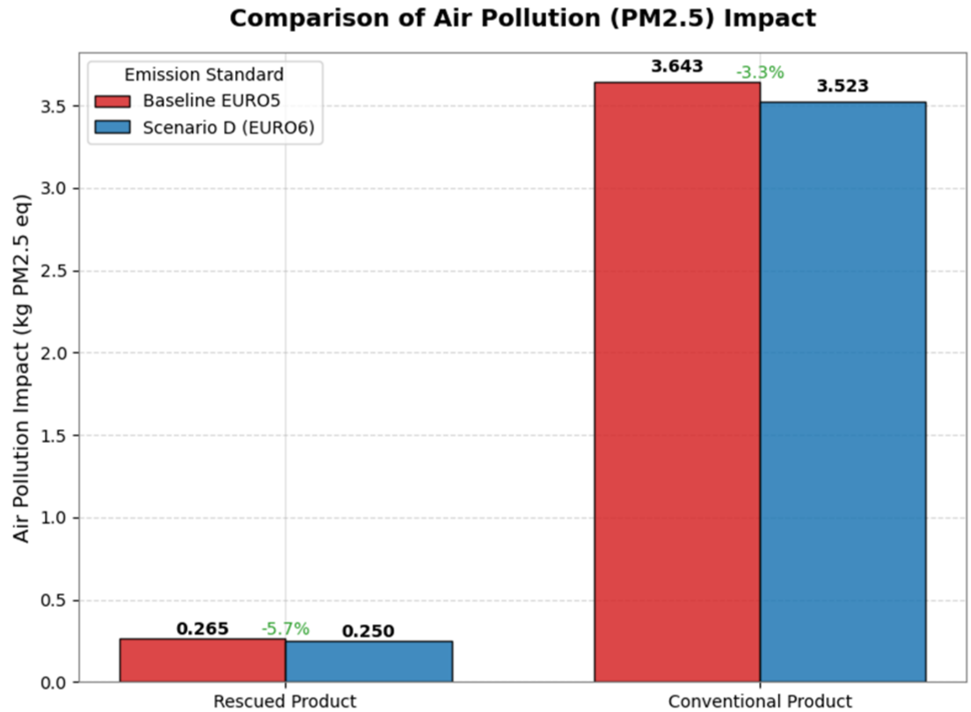

Figure 13 - Comparison of air pollution (PM2.5) after changing to EURO6 emissions standards

6. Conclusion

The life cycle assessment study investigates the comparison between the rescued bread flour produced by Rescued NZ and conventional wheat flour imported from Australia, following a cradle-to-gate approach. A particular focus of the study is the relevance of the carbon footprint of the rescued bread flour production process. The data used for the study relied on primary data provided by Rescued NZ, publicly available data through scientific literature and report, and lastly reliance on ecoinvent 3.8 database for all the wheat flour processes. The goal and scope of study framed the intended results to highlight: (1) the performance of the rescued bread flour compared to wheat flour, (2) the environmental hotspot from the rescued bread flour processes, and (3) the provision of robust and credible evidence of the potential environmental impact of rescued bread flour to Rescued NZ.

The study compares the two flours potential environmental impact based on the supplying 1 tonne of baking flour for baking purposes, in the Auckland region (NZ), stored in ambient room temperature. The study assessed five impact categories: carbon footprint, land use, freshwater ecotoxicity, fossil fuel use and air pollution (PM 2.5).

In summary, the study highlights the following findings:

a. The rescued bread flour outperforms the wheat flour in every impact category assessed in this study. This result is consistent throughout the iterations and uncertainty analysis. This is explained by the absence of intensive extraction of raw material; the surplus bread production is not accounted for the calculation. The extraction of raw material for wheat flour is rather intensive from farming processes and associated land use (wheat flour represents 99% of the land use category). Moreover, the rescued bread flour benefits from shorter transport distance as compared to the importation of wheat grain from Australia to New Zealand.

b. The study found that the drying is the main contributor to the rescued bread flour production processes, responsible for 82% of the total carbon emission, and 84% of fossil fuel use. This is attributed in part from the use of LPG gas (120 kW), and from the high electric consumption (6.5kW). A scenario has tested the used of energy produced by the electric boiler and it showed a decrease of more than 50% in carbon footprint and 41% in fossil fuel use.

c. The second main contributor to impacts is the paper packaging for the rescued bread flour, accounting for almost 87% of the land use. This is the result of forestry activity to produce the paper. The study explored a scenario where the paper bag is reused for 10 times. The results show an improvement of approximately 78% in land use, along with reductions across all other impact indicators due to lower resource extraction and energy requirements in production.

The study can conclude the superiority of the rescued bread flour over wheat flour with high certainty. Rescued NZ production system is an innovative solution to reduce carbon emission in NZ and food waste.

6.1 Limitations

It is clear from the results that the rescued bread flour, cradle-to-gate approach performs better in all impact categories, although those results rely on a few assumptions that can affect its overall precision. Firstly, the study considers the industrially made surplus bread to be the raw material, avoiding the impact from its production. This omission removes several steps that affects the environment, such as extraction of raw materials of all the ingredients, the manufacture's energy consumption and transport. Secondly, the wheat flour is imported from Australia, which includes heavy transportation. Lastly, the composition of the bread is unknown, which affects the reliability and the certainty of the modelling.

Moreover, the study focuses on a hybrid version of the current production (e.g., use of freezer) and future upscale scenario (e.g., use of flash spiral dryer), which is confusing to provide accurate data. Additionally, the study does not present the end-of-life phase, which measures the landfill impact of the wasted surplus bread. This would have highlighted the impact of the rescued bread flour environmental performance even further. This is a missed opportunity to explore as it is out of scope for the study. Due to time limit constraints, the study has not performed these alternatives scope. Finally, an LCA study always have a part of uncertainties, practitioner's mistake, even though the consistency and completeness check have been undertaken.

6.2 Recommendations

In accordance with the findings of the study, the environmental performance of the rescued bread flour could be improved by implementing alternative solutions, including:

a. Reusing the paper sack as often as possible; this will remove the burden of forestry activity. From the last discussion with Royce, it seems challenging to organise the tracking and collection of the used paper sack from customers. The logistic and the cost associated with the procedure do not compensate the effort, and the disposal of the paper sack seems to be more convenient. Hence this is a hotspot that should be investigated as the packaging represents the second main contributor to the environmental impact. A Life Cycle Cost (LCC) analysis could be done to assess the process and benefits.
b. Rescued NZ's efforts to rescued food waste could be highlighted by conducting a cradle-to-grave approach comparison with the wheat flour to explore the consequences of the use-phase, with different recipe ratios in baking cakes or bread, or a comparison with wasted surplus bread to evaluate the impacts of landfill.
c. By electrifying the FSD could be complemented with an LCC analysis to evaluate the financial feasibility and benefits in the long term.
d. A change of energy supplier to provide 100% renewable energy, which will significantly reduce the carbon footprint of the machines used.

The limitations and recommendations highlight the gaps in the study that could benefit for an improved performance of the rescued bread flour.

Thank you

6M Sustainability Consultants would like to thank Diane and Royce for providing the information for the study, and Febelyn Reguyal for her guidance throughout this process.

6M Sustainability Consultants
From left to right: Steven – Buna – Haochen – Cheng – Maharsi – Manja

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APPENDIX B. LCI results

1. Rescued bread flour

Methodology used: ReCiPe 2016 Endpoint (H)

Category	Impact Assessment result	Unit
Fine particulate matter formation	0.264584964	kg PM2.5 eq
Fossil resource scarcity	106.1339387	kg oil eq
Freshwater ecotoxicity	11.79156925	kg 1,4-DCB
Freshwater eutrophication	0.048537775	kg P eq
Global warming	405.3272409	kg CO2 eq
Human carcinogenic toxicity	10.82821483	kg 1,4-DCB
Human non-carcinogenic toxicity	148.6397188	kg 1,4-DCB
Ionizing radiation	4.250275758	kBq Co-60 eq
Land use	15.81053712	m2a crop eq
Marine ecotoxicity	15.18124561	kg 1,4-DCB
Marine eutrophication	0.005568815	kg N eq
Mineral resource scarcity	0.580344614	kg Cu eq
Ozone formation, Human health	0.461945907	kg NOx eq
Ozone formation, Terrestrial ecosystems	0.478810731	kg NOx eq
Stratospheric ozone depletion	9.51E-05	kg CFC11 eq
Terrestrial acidification	0.54358711	kg SO2 eq
Terrestrial ecotoxicity	890.1638733	kg 1,4-DCB
Water consumption	0.515197573	m3
Fine particulate matter formation	1.66E-04	DALY
Fossil resource scarcity	36.46358838	USD2013
Freshwater ecotoxicity	8.16E-09	species.yr
Freshwater eutrophication	3.25E-08	species.yr
Global warming, Freshwater ecosystems	3.10E-11	species.yr
Global warming, Human health	3.76E-04	DALY
Global warming, Terrestrial ecosystems	1.14E-06	species.yr
Human carcinogenic toxicity	3.59E-05	DALY
Human non-carcinogenic toxicity	3.39E-05	DALY
Ionizing radiation	3.61E-08	DALY
Land use	1.40E-07	species.yr
Marine ecotoxicity	1.60E-09	species.yr
Marine eutrophication	9.47E-12	species.yr

Mineral resource scarcity	0.134131529	USD2013
Ozone formation, Human health	4.20E-07	DALY
Ozone formation, Terrestrial ecosystems	6.18E-08	species.yr
Stratospheric ozone depletion	5.04E-08	DALY
Terrestrial acidification	1.15E-07	species.yr
Terrestrial ecotoxicity	1.02E-08	species.yr
Water consumption, Aquatic ecosystems	3.11E-13	species.yr
Water consumption, Human health	1.14E-06	DALY
Water consumption, Terrestrial ecosystem	6.96E-09	species.yr
End Point Classification		
DALY		
USD2013		
species.yr		

1. Wheat flour

Methodology used: ReCiPe 2016 Endpoint (H)

Name	Impact assessment result	Unit
Fine particulate matter formation	3.642958791	kg PM2.5 eq
Fossil resource scarcity	142.8740871	kg oil eq
Freshwater ecotoxicity	25.50815867	kg 1,4-DCB
Freshwater eutrophication	0.338883266	kg P eq
Global warming	1774.021735	kg CO2 eq
Human carcinogenic toxicity	29.37432476	kg 1,4-DCB
Human non-carcinogenic toxicity	394.4035624	kg 1,4-DCB
Ionizing radiation	9.568730674	kBq Co-60 eq
Land use	5957.021686	m2a crop eq
Marine ecotoxicity	33.642093	kg 1,4-DCB
Marine eutrophication	4.018737842	kg N eq
Mineral resource scarcity	6.953143616	kg Cu eq
Ozone formation, Human health	3.689424842	kg NOx eq
Ozone formation, Terrestrial ecosystems	3.778374733	kg NOx eq
Stratospheric ozone depletion	0.011395046	kg CFC11 eq
Terrestrial acidification	20.19844961	kg SO2 eq
Terrestrial ecotoxicity	3398.285758	kg 1,4-DCB
Water consumption	12.18492352	m3

Name	Impact assessment result	Unit
Fine particulate matter formation	0.002291306	DALY
Fossil resource scarcity	48.38260188	USD2013
Freshwater ecotoxicity	1.77E-08	species.yr
Freshwater eutrophication	2.27E-07	species.yr
Global warming, Freshwater ecosystems	1.36E-10	species.yr
Global warming, Human health	0.001646758	DALY
Global warming, Terrestrial ecosystems	4.97E-06	species.yr
Human carcinogenic toxicity	9.75E-05	DALY
Human non-carcinogenic toxicity	8.99E-05	DALY
Ionizing radiation	8.12E-08	DALY
Land use	5.29E-05	species.yr
Marine ecotoxicity	3.54E-09	species.yr
Marine eutrophication	6.83E-09	species.yr
Mineral resource scarcity	1.607136849	USD2013
Ozone formation, Human health	3.36E-06	DALY
Ozone formation, Terrestrial ecosystems	4.87E-07	species.yr
Stratospheric ozone depletion	6.05E-06	DALY
Terrestrial acidification	4.29E-06	species.yr
Terrestrial ecotoxicity	3.88E-08	species.yr
Water consumption, Aquatic ecosystems	7.36E-12	species.yr
Water consumption, Human health	2.71E-05	DALY
Water consumption, Terrestrial ecosystem	1.64E-07	species.yr
End Point Classification		
DALY		
USD2013		
species.yr		

APPENDIX D. Sensitivity analysis

1. Functional Unit

The Functional Unit (FU) is sensitive to the mass variance. The FU has been selected as it defines the core of the study. The study examines the changes in production mass when it increases or decreases by testing the sensitivity for FU = 0.5 tonnes, FU = 1 tonne, and FU = 1.5 tonnes.

As shown in Figure A, linear equal spacing can be observed across the three different FU parameters. This means that the change affects all computations in the model equally, thereby establishing a linear relationship between FU and the selected environmental impacts. Additionally, all five impact categories respond similarly when the FU is altered, showing that they are equally sensitive. The FU is not a sensitive parameter for those impact categories.

Figure B illustrate the carbon footprint from both flours. It shows the changes in parameters for the ***rescued bread flour*** (FU = 0.5 tonnes, FU = 1 tonne, and FU = 1.5 tonnes) and the baseline scenario for wheat flour (FU = 1 tonne). Interestingly, it shows that the ***rescued bread flour*** (FU = 1.5 tonne) produces just a third of the carbon footprint of wheat flour (FU = 1 tonne).

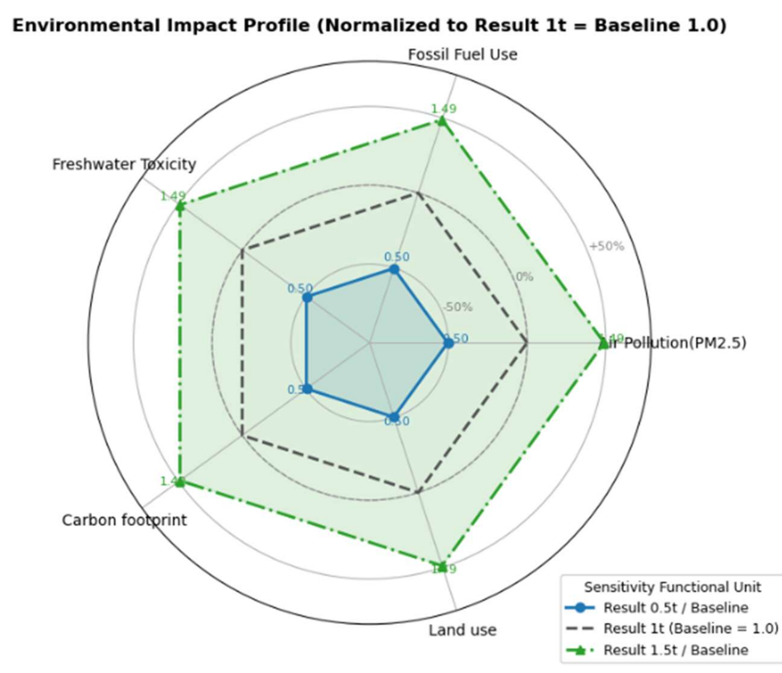


Figure A. Spiderchart Sensitivity Test of Functional Unit (mass tonne)

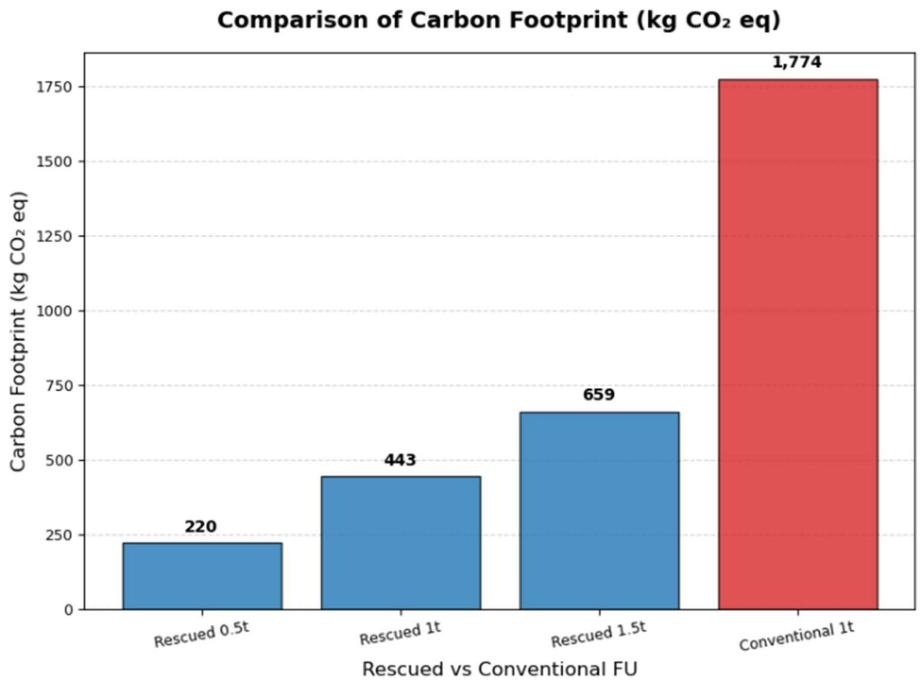


Figure B - Carbon Footprint comparison between Rescued FU and Wheat Flour FU

The study can conclude that the ***rescued bread flour*** performs better in terms of carbon emissions than the wheat flour, even with an increased production capacity.

2. Propane

Propane is one of the most common gases used in Liquefied Petroleum Gas (LPG). The modelling for the study defined propane in mass (kg) as part of the heat production unit process. The produced heat is utilised for the FSD process.

The sensitivity analysis investigates the effects on the impact categories of using more or less propane (e.g., +50% and -50%) for the FSD, assuming more or less efficient gas consumption. Figure C shows that the input of propane has a significant impact on fossil fuel use, compared to the FU analysis. This result is attributed to the nature of propane production. Propane is a byproduct of natural gas processing and crude oil refining (NREL, 2016) Hence, consuming more propane requires additional extraction of fossil fuels from oil and gas wells. On the other hand, changing the propane mass has less sensitive effects on land use, for instance. Figure D illustrates the comparison for fossil fuel use in the baseline scenario of 1 tonne of FU, showing a variance of 25.7% with wheat flour, as indicated in the results interpretation. With a 50% increase in propane use, the wheat flour actually performs better. This can be attributed to the significant environmental impact of propane production.

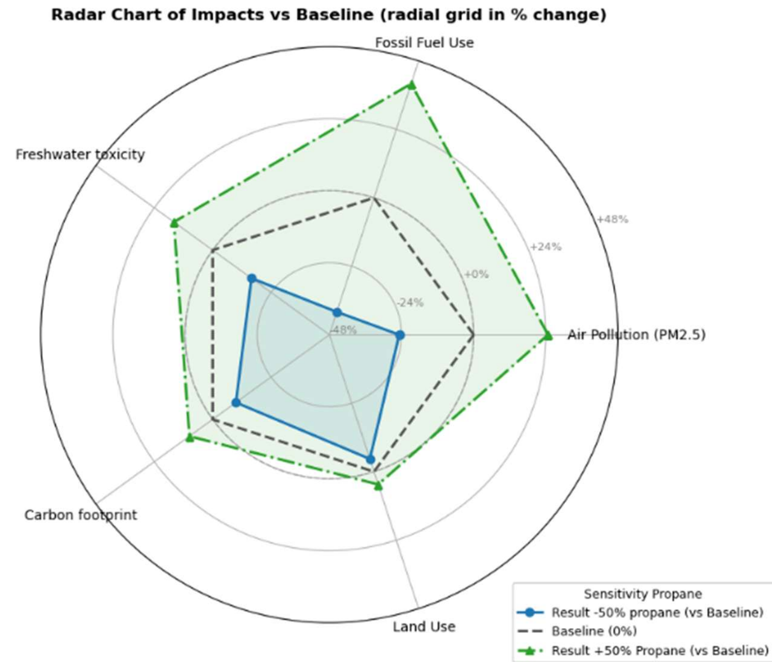


Figure C. Spider Chart Sensitivity Test of Propane (Mass kg)

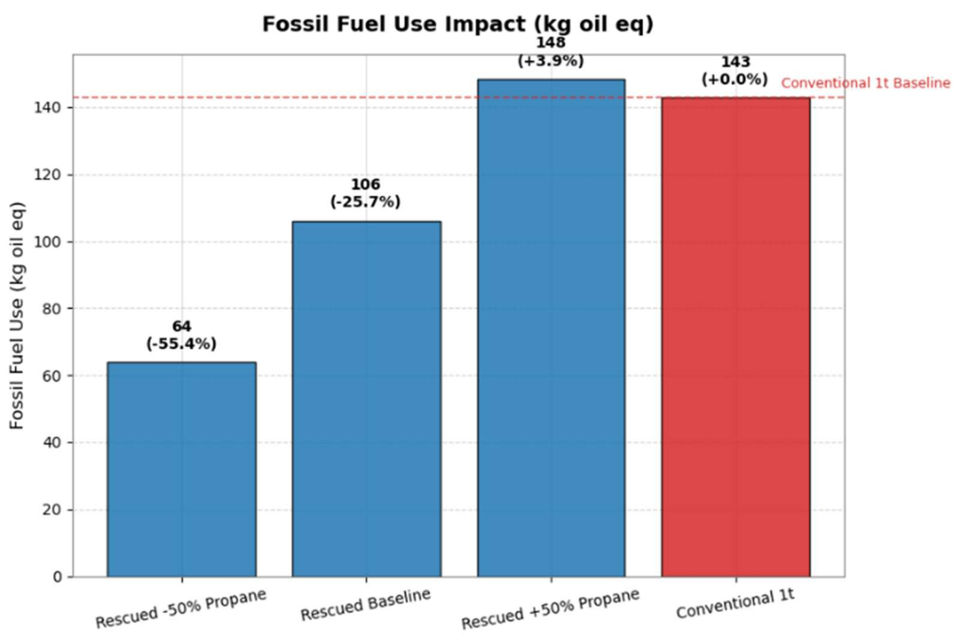


Figure D. Fossil Fuel Use Comparison between Rescued and Conventional Flour

3. Electrical Efficiency

Electrical efficiency is related to the electrical load, measured in kilowatt-hours (kWh), required to operate the equipment. Electricity production relies on the remarkable properties of copper. Copper has superior electrical and thermal conductivity compared to other materials (ICA, 2021) . It is widely used in low-voltage circuits, spanning from the distribution and electricity production chain at power plants to households. However, the production of copper and waste treatment processes, such as municipal incineration of copper scrap, increase the toxicity levels to freshwater (Norgate & Rankin, 2000) . Metal copper is among the most ubiquitous contaminants in freshwater aquatic environments; due to its widespread use, it greatly affects the freshwater biodiversity and ecosystem (e.g., fish, invertebrates, plants, and algae) at high concentrations (MfE, 2025) .

In the production line, the FSD equipment has the highest electrical load among all unit processes. Based on the **Rescued NZ** data, FSD equipment requires approximately 65 kWh to produce 1 tonne of ***rescued bread flour***. The study examined the change in electricity efficiency by assessing a system that is 50% more efficient, requiring fewer kWh. In practice, this can be achieved by investing in new technology. The study also tested a 50% reduction in efficiency, which is likely to happen due to equipment ageing or poor maintenance.

Figure E shows that the electricity efficiency is a sensible parameter for freshwater toxicity, especially with a 50% decrease in efficiency. Meanwhile, land use is less affected by changes in electrical efficiency.

Figure F illustrates the freshwater toxicity in the baseline scenario, where 1 tonne of FU is present. It is observed that the ***rescued bread flour*** causes less harm to freshwater toxicity compared to the wheat flour. This result is consistent even with a 50% decrease in efficiency.

The study can conclude that the energy consumption uses for the rescued bread flour is not targeted as an environmental hotspot, despite the high energy use of the FSD. This is partly due to the minimal use of heavy equipment used by **Rescued NZ.**

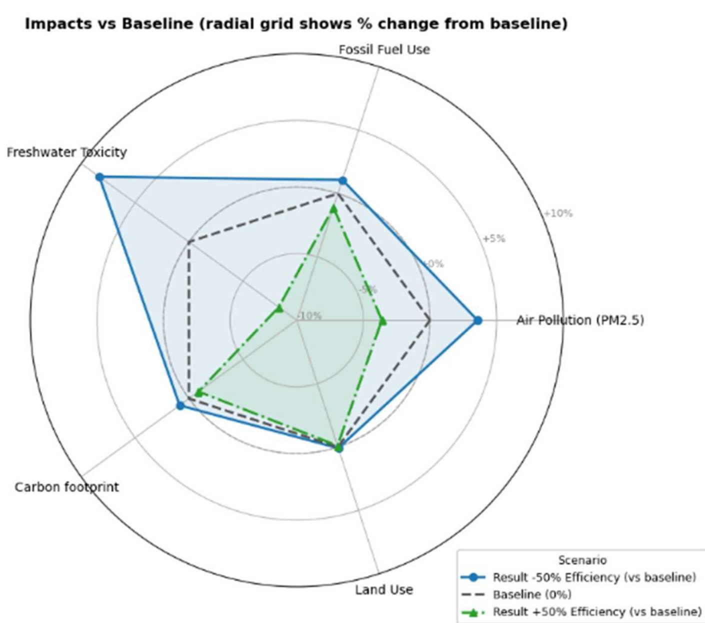


Figure E. Spider Chart Sensitivity Test of Electrical Efficiency (kilowatt-hour)

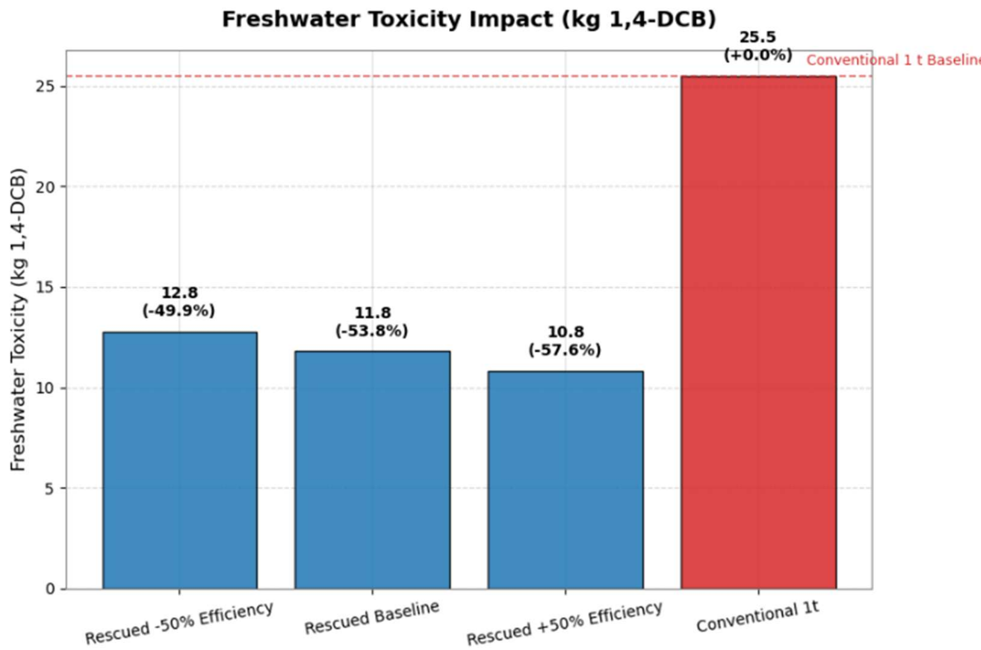


Figure F. Freshwater Toxicity Comparison between Rescued and Conventional Flour

Appendix E. Uncertainty analysis (Monte Carlo simulation)

1. Result consistency

Table 1. Uncertainty Analysis Results of Mid-point Indicators (Rescued Bread Flour)

Impact Category	Units	Deterministic Result	Mean	Std. Dev	5th Percentile	95th Percentile	90% Confidence Interval
Carbon footprint	kg CO_2 eq	405.33	405.00	9.81	390.43	421.88	[390.43, 421.88]
Fossil fuel use	kg oil eq	106.13	105.82	8.72	92.40	120.87	[92.40, 120.87]
Land use	m^2a crop eq	15.81	15.85	8.34	6.96	30.23	[6.96, 30.23]
Freshwater toxicity	kg 1,4-DCB	11.79	11.77	3.15	7.63	17.17	[7.63, 17.17]
Air pollution (PM2.5)	kg PM2.5 eq	0.26	0.26	0.02	0.23	0.30	[0.23, 0.30]

Table 2. Uncertainty Analysis Results of Mid-point Indicators (Conventional Wheat Flour)

Impact Category	Units	Deterministic Result	Mean	Std. Dev	5th Percentile	95th Percentile	90% Confidence Interval
Carbon footprint	kg CO_2 eq	1786.76	1780.27	207.63	1446.07	2149.86	[1446.07, 2149.86]
Fossil fuel use	kg oil eq	143.82	143.40	23.14	109.75	184.47	[109.75, 184.47]
Land use	m^2a crop eq	6000.78	5999.72	852.89	4695.58	7520.82	[4695.58, 7520.82]
Freshwater toxicity	kg 1,4-DCB	25.68	25.45	9.27	12.09	41.99	[12.09, 41.99]
Air pollution (PM2.5)	kg PM2.5 eq	3.67	3.66	0.55	2.86	4.59	[2.86, 4.59]

For both ***rescued bread flour*** and conventional wheat flour, the deterministic results are very close to the mean values obtained from Monte Carlo simulation. For example, in the rescued flour system, the deterministic carbon footprint is 405.33 kg CO_2 eq, while the simulated mean is 405.00 kg CO_2 eq. The small difference (< 0.1%) suggests that the model is well balanced and not dominated by outlier parameters. A similar trend is observed for all other indicators, indicating that the deterministic results can be considered representative of the mean system behaviour.

2 . Coefficient of variance

Table 3. Comparison of Coefficient of Variance between two flours

Impact Category	Rescued Bread Flour	Conventional Wheat Flour
Carbon footprint	2.4%	11.7%
Fossil fuel use	8.2%	16.1%
Land use	52.6%	14.2%
Freshwater toxicity	26.8%	36.4%
Air pollution (PM2.5)	8.6%	14.9%

The coefficient of variation (CV = Std. Dev / Mean) was used to measure the relative uncertainty of each indicator. Rescued flour shows low relative uncertainty for carbon footprint, fossil fuel use, and PM2.5 (CV < 10%), moderate for freshwater toxicity, and high relative uncertainty for land use. The relatively high CV of land use (52.6%) for rescued bread flour is mainly driven by the packaging stage, where data were based on generic database and assumptions for paper bag production. Such secondary data often have higher uncertainty ranges due to regional and technological variability.

For the conventional wheat flour, higher uncertainty in all categories reflects the inherent variability in agricultural dataset .

3. Variability and confidence intervals (similar with former part)

The width of the 90% confidence intervals reflects the uncertainty associated with each impact category.

For rescued bread flour, all indicators show relatively narrow ranges — e.g., carbon footprint varies between 390 and 422 kg CO_2 eq (±2.4%), fossil fuel use between 92 and 121 kg oil eq (±8.2%), and air pollution (PM2.5) between 0.23 and 0.30 kg PM2.5 eq. This indicates low model uncertainty and high result reliability.

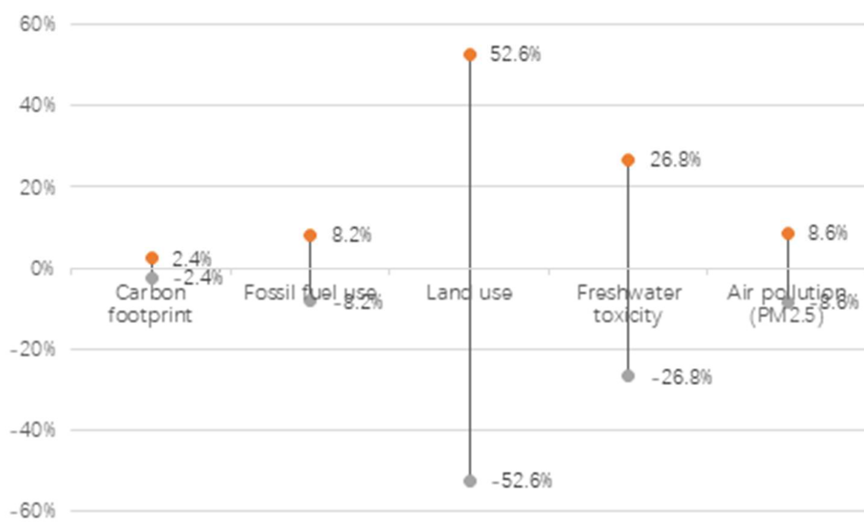


Figure 1. Uncertainty analysis for rescued bread flour

In contrast, conventional wheat flour shows considerably wider confidence intervals. Its carbon footprint, for instance, ranges from 1,446 to 2,150 kg CO_2 eq (±11.7%), which is already the least uncertain among its impact categories. Such larger deviations suggest that the conventional wheat flour system has higher uncertainty, mainly because its underlying life cycle data are sourced from generic database inventories rather than primary or site-specific measurements.

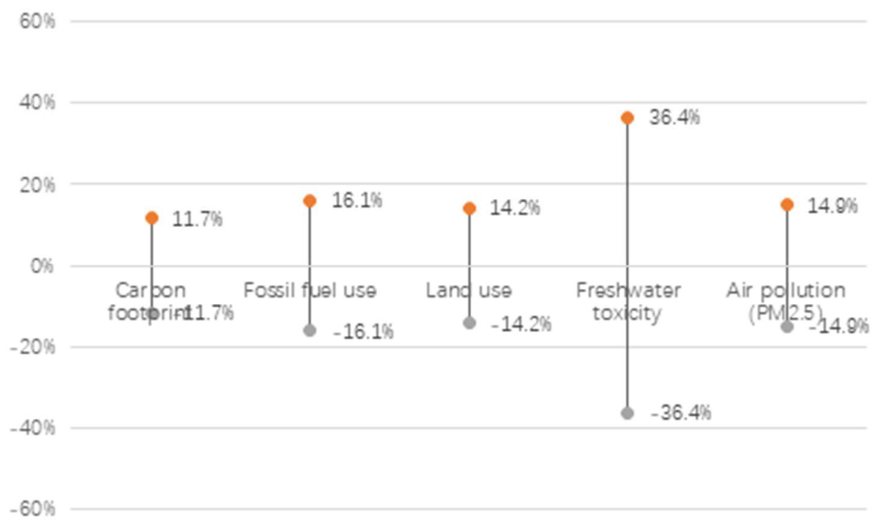
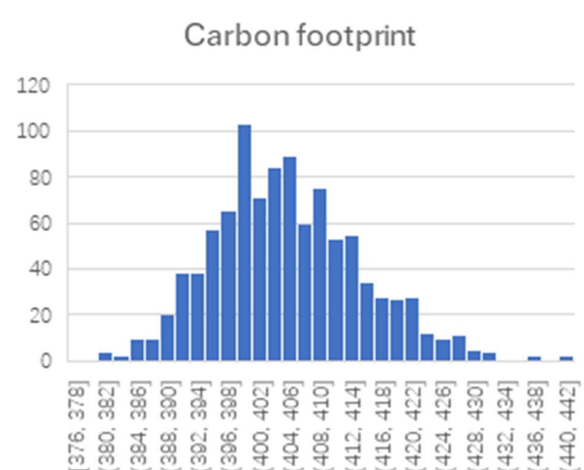


Figure 2. Uncertainty analysis for conventional wheat flour

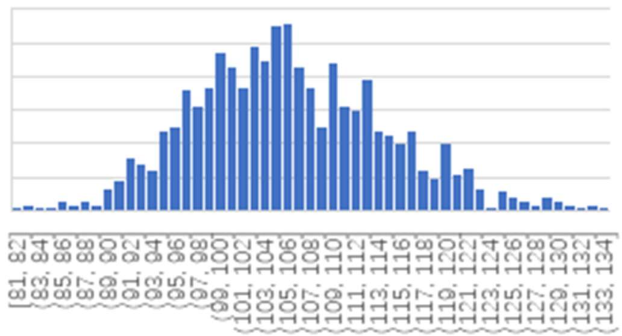
4. Key implications

The low standard deviation and tight confidence intervals of the rescued flour system indicate strong model reliability. The comparatively high variability of conventional wheat flour highlights the inherent uncertainty of agricultural systems. The comparative advantage of rescued bread flour remains consistent across the Monte Carlo trials, confirming that waste valorisation significantly reduces environmental impact even when uncertainty is considered.

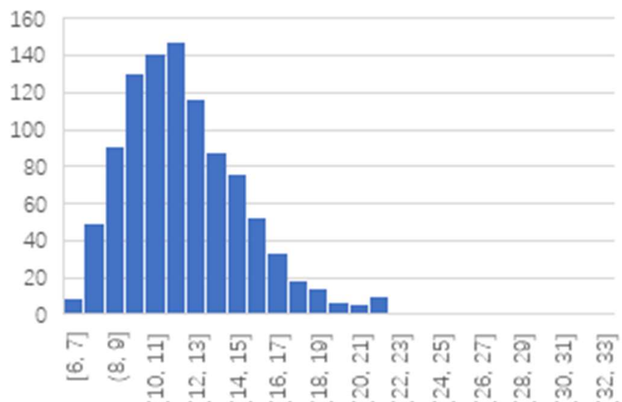
5. Distribution of Monte Carlo Simulation of Rescued Bread Flour (Blue graphs)



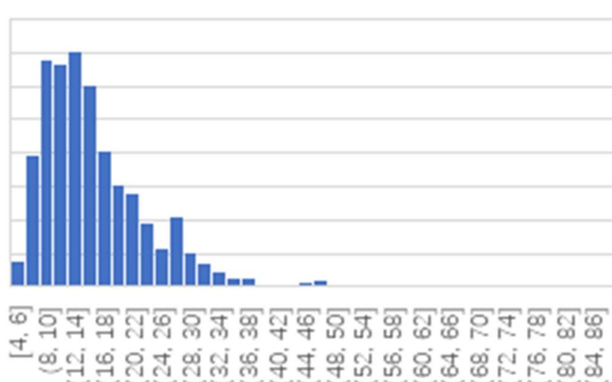
Fossil fuel use

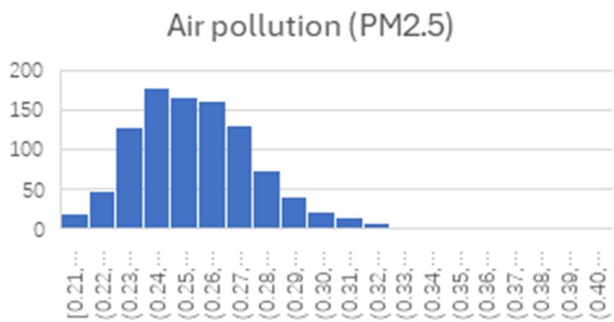


Freshwater toxicity

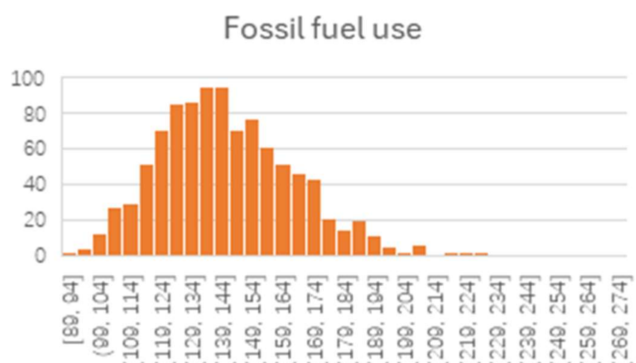


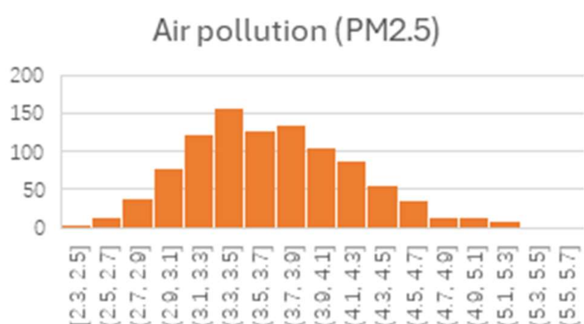
Land use

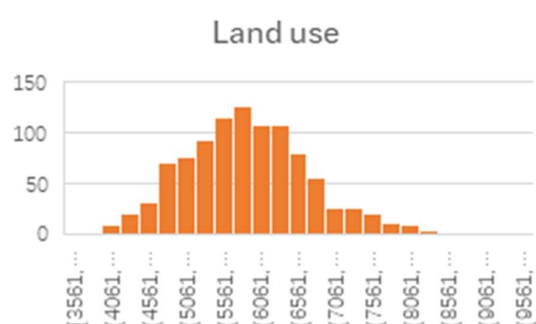


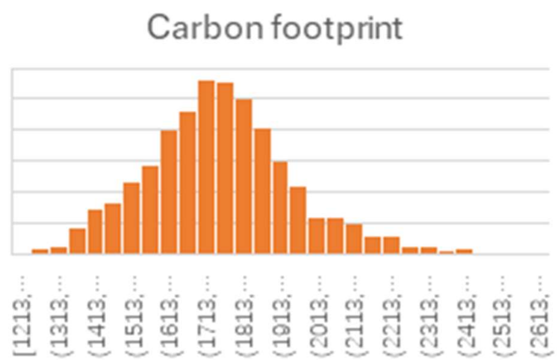


6. Distribution of Monte Carlo Simulation of Conventional Wheat Flour (Orange graphs)








Carbon footprint
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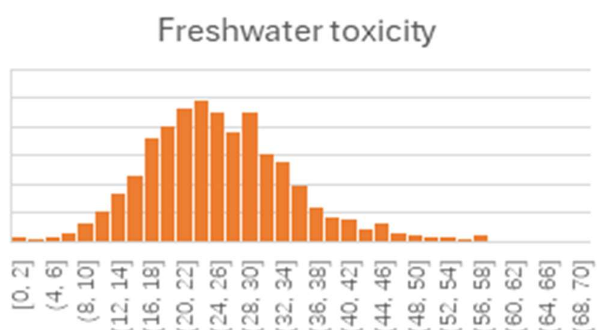
Freshwater toxicity
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(4, 6]
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(12, 14]
(16, 18]
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(36, 38]
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(44, 46]
(48, 50]
(52, 54]
(56, 58]
(60, 62]
(64, 66]
(68, 70]

Appendix F. Completeness and consistency checklist

ID	Check Item	Rescued Bread Flour	Wheat Flour	Notes/Evidence
CROSS CUTTING CONSISTENCY				
1	Study follows ISO 14044 standards	Yes	Yes	
2	Terminology used consistently	Yes	Yes	The terms and its explanation were defined for general audiences
3	Aggregation level justified and consistent	Yes	Yes	Aggregation level of wheat flour productions is aggregated into one process of wheat grain processing in ecoinvent
4	Value choices documented and evaluated	Yes	Yes	
FUNCTIONAL UNIT CONSISTENCY				
1	Identical functional unit clearly defined	Yes	Yes	
2	Functional properties equivalent	Yes	Yes	
3	Main product clearly defined	Yes	Yes	
4	Reference flow defined consistently	Yes	Yes	Using surplus bread as input for rescued bread flour and wheat grain as input for wheat flour (raw material)
SYSTEM BOUNDARY CONSISTENCY				
1	System boundary defined as cradle to gate for both	Yes	Yes	EoL is excluded for both SB
2	Geographic boundary documented	Yes	Yes	
3	Temporal boundary documented	Yes	Yes	
4	Technological boundary documented	Yes	Yes	
5	Same life cycle stages included	Yes	Yes	from raw material extraction to packaged at warehouse, using same end point at Rescued
6	Transportation included consistently	Yes	Yes	

7	Packaging included	Only for final product	Only for final product	
8	End-of-life excluded and justified	Yes	Yes	Assumed the EoL after use phase is similar for both products
IMPACT ASSESSMENT CONSISTENCY				
1	Same LCIA Method used	Yes	Yes	
2	Same impact categories assessed	Yes	Yes	
3	Same characterization factors applied	?	?	
DATA CONSISTENCY				
1	Same database version used	Yes	Yes	Ecoinvent 3.8 or 3.11? How to check the version on open LCA??
2	Data quality requirement equivalent	?	?	Waiting for pedigree analysis
3	Geographic representativeness documented	Yes	Yes	
4	Temporal representativeness documented	Yes	Yes	
5	Technological representativeness documented	Yes	Yes	
6	Uncertainty estimations documented	Yes	Yes	Yes, on the goal and scope definition
ALLOCATION CONSISTENCY				
1	Allocation procedures documented	Yes	N/A	Using allocation to divide energy use and emission from sieving to make bread flour and breadcrumbs
2	Same allocation method applied	Yes	Yes	
3	Byproduct handling consistent	Yes	Yes	Exclude breadcrumbs from SB
MODELLING APPROACH CONSISTENCY				
1	Modeling approach documented	Yes	Yes	Using attributional approach
2	Attributional approach applied consistently	Yes	Yes	
3	No overlap of unit processes	Yes	Yes	

4	Calculation procedures identical	Yes	Yes	Using "kg" of as unit instead of "tonne" for LCI collection and modelling, multiply to tonne in LCIA phase
ELECTRICITY MODELING CONSISTENCY				
1	Voltage level selection documented	Yes	Yes	Using NZ electricity mix (own model) or using market for electricity?
2	Same voltage level used for similar operations	Yes	Yes	High? Medium? Low?
3	Electricity mixed justified	Yes	Yes	Using 88% of RE generation with justified RE sources
TRANSPORTATION MODELING CONSISTENCY				
1	Transport mode documented	Yes	Yes	
2	Transport distances documented	Yes	Yes	
3	Same transport dataset type used	Yes	Yes	
4	Load factors documented	Yes	Yes	
5	International shipping included (if applicable)	N/A	Yes	International shipping from AUS to NZ port
CUT-OFF CRITERIA CONSISTENCY				
1	Same mass cut-off applied	Yes	Yes	
2	Same energy cut-off applied	Yes	Yes	
3	Same impact cut-off applied	Yes	Yes	
4	Exclusions applied consistently	Yes	Yes	
INTERPRETATION CONSISTENCY				
1	Same contribution analysis method	Yes	Yes	
2	Same level of result aggregation	Yes	Yes	
3	Results presented in same format	Yes	Yes	What chart/table/graphs should we use?
4	Same sensitivity parameters tested	Yes	Yes	
5	Same uncertainty methods applied	Yes	Yes	